FREDERIC KORAN

Engineering Project Portfolio

Hi, I'm Fred Koran, a fourth-year student in McGill's five-year Bioengineering program, working in the Biomolecular and Cellular Engineering stream while pursuing a minor in Economics.

I'm also a two-sport varsity athlete, returning to the varsity lacrosse and varsity track and field teams at McGill this fall, after running track and playing soccer through high school. Between terms I spent three summers building and running a landscaping business of my own and a season coaching high school track, and I'm now in product development at Ruffian, a student-run startup building the TipTopSpin tennis trainer.

What follows is an overview of the significant academic projects I've worked through: device design, biomaterials, modelling and laboratory measurement.



INSIDE

Engineering projects in diagnostic device design, biomaterials, structural biology, biosignal analysis, nanomaterial synthesis and analytical measurement.

PROGRAM

B.Eng. Bioengineering, Biomolecular & Cellular Engineering. Minor in Economics. Expected May 2028.

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01 · DEVICE DESIGN STUDY · WINTER 2026 · FOUR-PERSON TEAM

Channel Geometry as a Design Variable in a Sickle Cell Point-of-Care Device

Four devices differing only in shape moved aggregation onset from 17% to 64% of channel length. The variable costs nothing per unit.

Problem

Microfluidic point-of-care tests for sickle cell disease fail through a specific mechanism. Rigid cells aggregate and partially occlude the channel, fluid appears to keep flowing, and the readout then describes the device rather than the patient.

Approach

A 2 × 2 matrix of channel geometries: serpentine against spiral path, square against U cross-section. All four were 3D printed and tested under matched fluid, loading and material conditions with a blood-mimicking cell suspension.

Result

Aggregation onset ranged from 17% to 64% of channel length across four devices differing only in shape. The geometry the literature predicted would perform best clogged first.

WHY GEOMETRY

Sickle cell disease affects roughly 7.74 million people, with about 515,000 new births a year and 81,100 deaths in children under five. Around 80% of cases occur in sub-Saharan Africa, where confirmatory electrophoresis and HPLC are unavailable at the point of care. The point-of-care assays that are available cost $1.40 to $4.75 per test and remain qualitative. Geometry was selected as the design variable because it is the only one that is free. Reagents cost money, coatings cost money, instrumentation costs money. Shape changes device behaviour and adds nothing to the unit cost.

BUILD AND TEST CONDITIONS

Channels were 1 mm square, with a 0.5 mm radius on the U profiles, printed in 405 nm resin, ethanol-washed and sealed with adhesive film. The base fluid was 1% BSA at 10 mg/mL in pH 7.4 PBS, selected over Pluronic F127 so that protein fouling would be reproduced rather than suppressed. BSA alone fails as an aggregate surrogate: at 7 to 10 nm it is roughly a thousand times smaller than the 5 to 10 µm cells it stands in for. *Saccharomyces cerevisiae*, cultured 24 h at 30 °C and mixed 55:45 with the base fluid, supplied a cell-scale surrogate at approximately 1.6×10^7 cells/mL by OD600. That sits at the low end of the range used in microfluidic blood models, at comparable cell diameter.

All four geometries carried the cell-free control the full length of the channel. Under load they separated sharply.

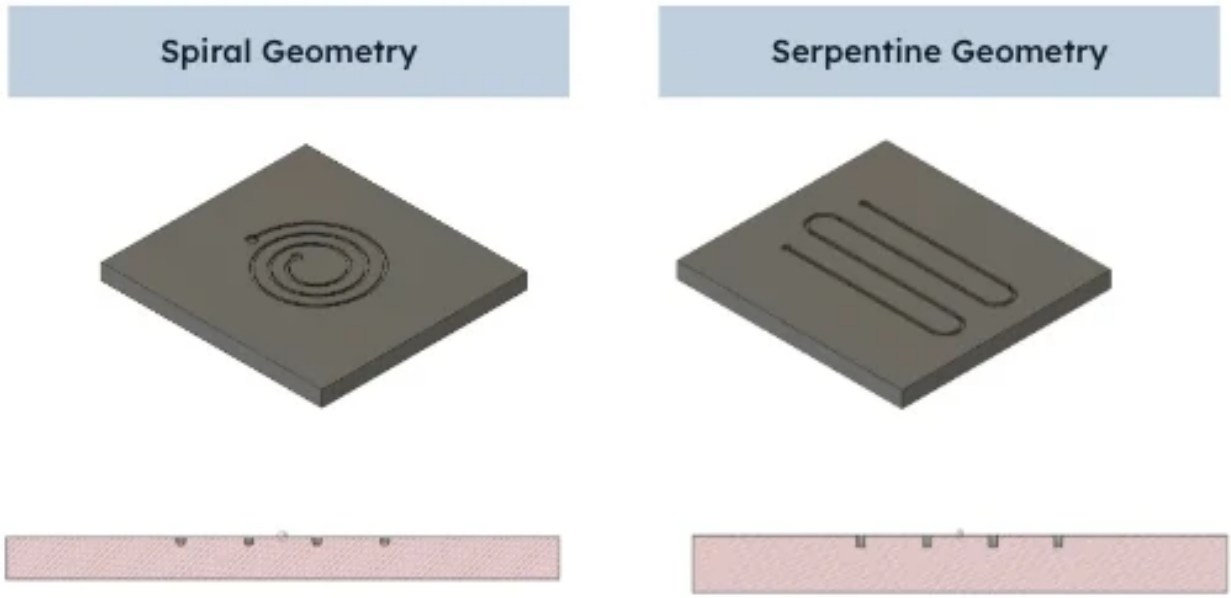


Figure 1.1 The four devices, as modelled. Spiral against serpentine path, and the 1 mm square channel against the 0.5 mm radius U cross-section common to all of them. Nothing else differs between the four.

THE INVERTED HYPOTHESIS

The hypothesis was spiral path with U cross-section, on the expectation that Dean vortices would break aggregates apart, which is what the literature predicts. It came out inverted. That geometry produced the highest flow rate of the four and clogged after 20% of the channel, while serpentine-square travelled three times further. The mechanism is as follows. Serpentine channels generate inertial focusing that drives particles off the wall, and square cross-sections produce symmetric flow. The spiral-U combination generates a Dean number high enough that continuous rotational mixing promotes aggregate contact instead of preventing it. Mixing intensity is not an unqualified advantage. Flow stability, rather than flow speed, carried fluid further.

LIMITS AND NEXT ITERATION

Each condition is a single run, with no replicates, no error bars and no statistics, in an open rather than closed device, loaded by micropipette so applied pressure varied between runs. The result supports a rank ordering of four geometries under matched conditions. It does not constitute a validated device. The 17% to 64% spread is wide enough to survive the sparsity of the data; the ranking of the middle two geometries is not.

Replace the pipette with a syringe pump to remove the pressure variable, add replicate runs, substitute erythrocytes for yeast to capture non-Newtonian behaviour, and move from resin to thermoplastic for surface quality and scale. All four requirements came directly out of the fabrication study in project 02.

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Low-Cost Microfluidic Device Fabrication with PDMS Shrinky Dink Replication and LCD Stereolithography

A 0.1 mm channel would not form and a 95.9° contact angle removed capillary flow: two hard process limits, and a result held at inconclusive rather than reported as a finding.

Microfluidic performance is set by fabrication method, channel geometry and surface wettability together. The two accessible routes to a channel were run side by side under matched conditions: PDMS cast against a shrunk-thermoplastic master, the Shrinky Dink method, and direct LCD stereolithography in 405 nm resin.

ROUTE ONE: SHRUNK THERMOPLASTIC MASTER

A pre-stressed thermoplastic sheet was measured, baked at 160 °C for five minutes and re-measured. The contraction factor came out at 0.45. In-plane shrinkage raises a printed pattern into relief, so channel walls form without a cleanroom. Patterns were printed, shrunk, and the master cut to size by that factor before PDMS at a 1:10 curing-agent ratio was degassed, poured over it and cured at 160 °C for an hour. Inlets and outlets were punched and the device sealed under press-on glass.

The channels sealed cleanly: no leaks, no trapped bubbles, no feature collapse. That is consistent with published shrink factors near 0.4, and with collapse being reported only below roughly 200 µm, well under the millimetre-scale features here. Wetting is what failed. A droplet placed at the inlet did not cross the channel. Ethanol in the test liquid improved wettability, but pipette pressure was still required to move fluid at all.

ROUTE TWO: LCD STEREOLITHOGRAPHY

Channels of increasing width were drawn in CAD, printed on an ELEGOO Mars in 405 nm resin, ethanol-washed and sealed with glass. Dimensional accuracy held everywhere except the smallest feature. Each channel was loaded with a fixed 50 µL and flow measured from video in ImageJ.

Channel width	Channel filled after 50 µL	Flow rate under applied pressure
0.1 mm	Channel did not form	—
0.5 mm	100%	8.3 mm/s
1.0 mm	50%	7.5 mm/s
2.5 mm	15%	2.5 mm/s
5.0 mm	2%	0.33 mm/s
1.0 mm (PDMS route)	100%	6.3 mm/s

The 0.1 mm channel failing to form sets the printable floor for this equipment, and it matches reports that sub-100 µm features require a specialized setup. The apparent fall in flow rate with rising width is confounded by design. Loading volume was fixed at 50 µL while channel volume grew, so wider channels never filled and their fronts stalled mid-channel. Rates of 0.33 to 8.3 mm/s under gentle pressure agree with published values for untreated polymer chips.

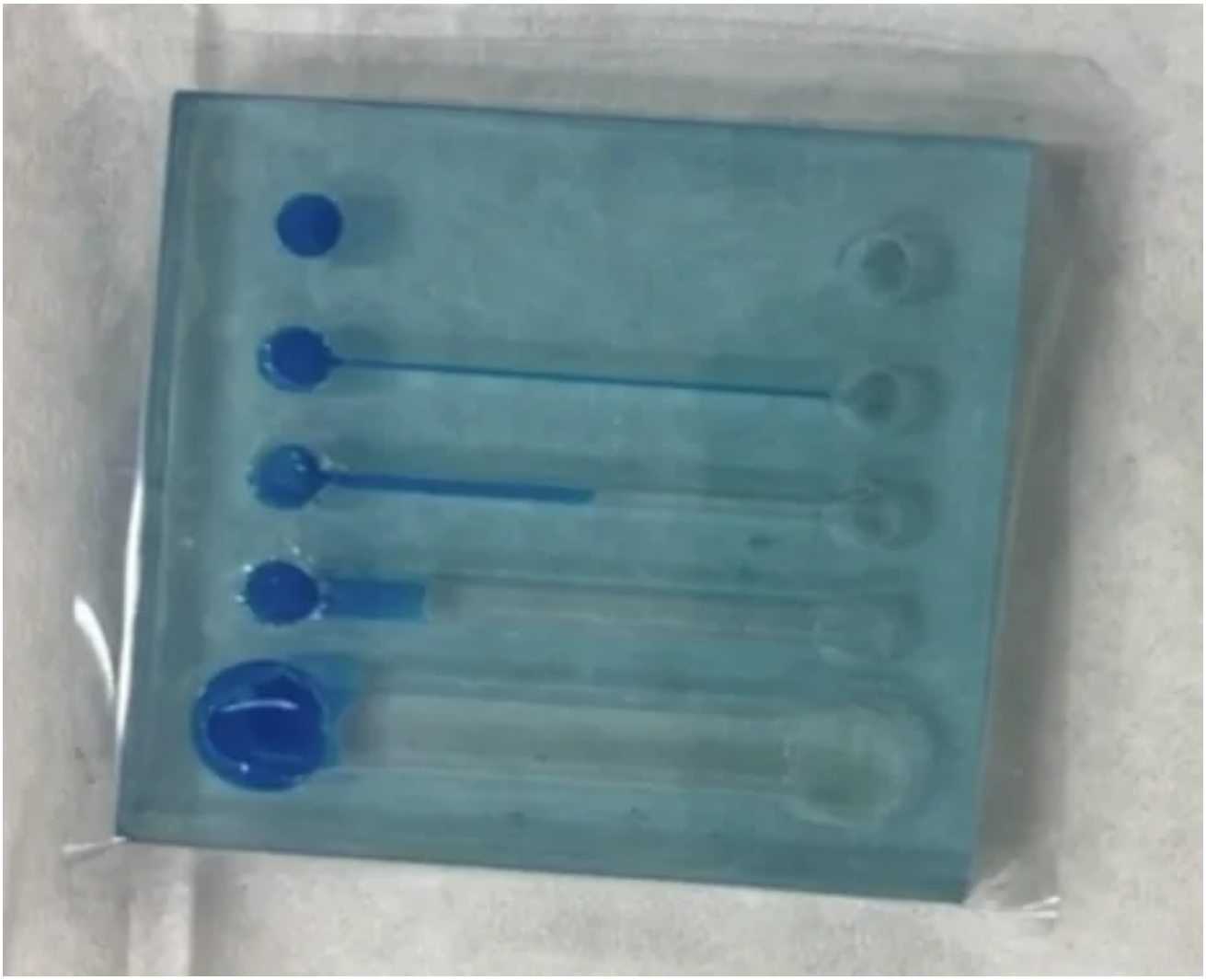


Figure 2.1 The printed straight-channel array after 50 µL of dye was loaded into each channel. Fill fraction drops as channel width increases, and the 0.1 mm channel did not form.

MIXING, AND WHY THERE WAS NONE

Two mixing geometries were printed with dual dye inlets and loaded with 50 µL each in a 25:75 ethanol to water dilution to improve wetting. Neither produced measurable mixing. The limiting factor was not miscibility but the failure to establish stable co-flow: without both streams filling the channel at once there is no interface across which diffusion can occur. This is the expected outcome. Co-current laminar streams without a dedicated mixing structure do not mix, and recovering mixing requires serpentine or herringbone patterning, grooves, or split-and-recombine elements, all of which still require the channel to fill.

A 95.9° contact angle measured directly on PDMS sits just above the 90 to 95° reported in the literature. At that angle capillary filling is negligible, which explains the whole set of observations at once: no spontaneous flow, pressure-driven transport only, and no stable interface for mixing.

VERDICT, AND WHAT IT SET DOWNSTREAM

The hypothesis was reported as inconclusive rather than rejected, deliberately. Fixed loading volume against varying channel volume, manual pipette pressure and hand loading left too much methodological uncertainty to support a conclusion about flow stability or mixing. Calling a null result a finding here would not have been supportable.

Three requirements came out of it: a pump instead of pipette pressure, a wettability treatment or a different material, and channels sized for measurable flow. Those are the requirements the device work in project 01 was built around.

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Commercial and Universal Lateral Flow Assays

The lab-assembled strip ran correctly and still returned false negatives. The control line establishes that the failure was chemistry rather than flow.

A lateral flow assay is a diagnostic reduced to a strip: a sample pad, a conjugate pad carrying antibody-labelled gold nanoparticles, a nitrocellulose membrane striped with capture antibodies, and an absorbent pad that pulls the sample across by capillary action. The user experience is nearly identical across products. The engineering underneath is not. Three commercial tests were run to protocol and then dismantled, and an assay was assembled from modular reagents and benchmarked against one of them on the same samples.

THREE PRODUCTS, THREE SAMPLE MATRICES

Assay	Sample matrix	Recognition	What the teardown showed
Pregnancy (hCG)	Urine	Sandwich immunoassay, AuNP reporter	Direct sample addition and readout; positive on pure and 8× diluted alike
Mononucleosis	Whole blood, serum, plasma	Bovine erythrocyte-derived antigen	Single-pad internal geometry, adapted to a blood matrix rather than to a different target
Streptococcus A	Throat swab extract	Antigen capture after extraction	Extraction step required before loading; same sandwich principle downstream

All three detected their analyte and all three formed a control line. What they share is a transport mechanism and little else. Sample handling, pad architecture and recognition chemistry are each re-engineered around the matrix the test has to accept, which is why a blood test and a swab test cannot be built on the same strip even when both detect a protein.

EXTENDING THE READOUT PAST THE EYE

Visual inspection is the design constraint of these devices and also their sensitivity ceiling. Gold nanoparticles absorb strongly in the green and convert that energy to heat, so the reporter line was excited with a green laser and imaged on an infrared camera. The thermal signature located the line independently of its visible colour. That is the principle behind thermal-contrast readers, and what it buys is sensitivity recovered from strips already manufactured and already in the field.

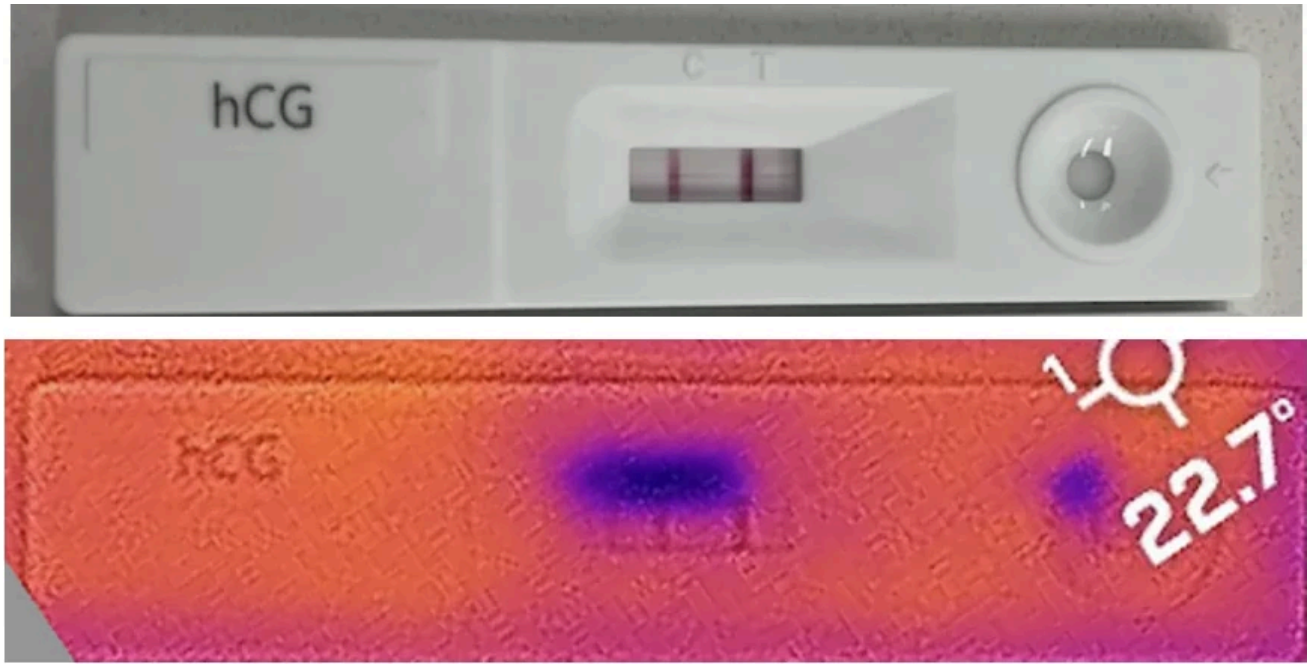


Figure 3.1 The same commercial hCG cassette twice. Read by eye at the top, and imaged on an infrared camera after green laser excitation of the gold nanoparticle reporter at the bottom. Both lines register thermally.

THE BENCHMARK

A universal lateral flow assay was assembled from modular conjugation reagents, incubated five minutes, and run against a commercial hCG cassette on the same two samples.

Device	Sample	Control	Test line	Interpretation
Commercial hCG cassette	Pure positive	Present	Present	Positive
Commercial hCG cassette	8× diluted	Present	Present	Positive
Lab-assembled strip	Pure positive	Present	Absent	False negative
Lab-assembled strip	8× diluted	Present	Absent	False negative

Because the commercial cassette read positive on both samples, hCG was present above a detectable threshold, so the assembled strip's negatives are false negatives rather than true ones.

The control line is the diagnostic detail. It appeared in both runs, which means sample was drawn the length of the membrane and the strip was mechanically sound. Capillary flow worked. Antigen recognition did not. That isolates the failure to conjugation chemistry and reagent preparation: incomplete or unstable antibody-nanoparticle conjugation, insufficient antibody concentration, or a buffer that failed to carry the complex to the test line. It also quantifies what a commercial device is actually selling. Not the strip and not the flow, but the antibody selection, conjugation chemistry and membrane treatment that a kit assembled by hand does not reproduce.

LIMITS

Two samples, one strip each, no replicates and no titration series, so the failure is located but not measured. There is no detection limit for the assembled assay because it never produced a signal. Separating conjugation efficiency from antibody concentration as the cause requires testing the conjugate on a dot blot before it is committed to a strip, which is the check this experiment lacked.

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Synthesis and Characterization of Functional Nanomaterials: AgNP-Diatom Composites, Protein Adhesives, and Carbon Nanodots

Four materials, five instruments, and two results that ran against the literature. One of them traces to the deposition rather than to the material.

Four materials were synthesized from scratch and then characterized against a single hypothesis each, so that every claim rested on an instrument reading rather than on appearance.

Material	Synthesis route	Characterization
Silver nanoparticles	Citrate reduction of 1 mM $AgNO_3$ at boiling, 30 to 60 min	UV-Vis absorbance, 300 to 800 nm
Carbon nanodots	Hydrothermal, 4% glucose at 200 °C for 6 h, centrifuged and syringe-filtered	Fluorescence excitation and emission scans
Silver-diatom biocomposite	0.25 g diatoms homogenized in AgNP solution, evaporated and oven-dried at 120 °C	Stereoscopy, cyclic voltammetry, contact angle
Egg albumin adhesive	Dried egg white ground and rehydrated at 0.875 g/mL, cured 30 min between leather	Tensile testing, contact angle

WHAT THE OPTICS CONFIRMED

Citrate reduction turned the solution from transparent yellow to opaque dark brown, and the plasmon peak sat at 435 nm in every silver-containing sample, which matches the published value. Peak absorbance rose with diatom loading, from 0.363 AU for pure AgNP to 0.387 AU at the lower loading and 0.422 AU at the higher. That is consistent with the silica frustules enhancing scattering near the particle surface, and it was confirmed independently on a second spectrometer. Carbon nanodots emitted at 370 nm at 23,254 a.u. against a flat water blank.

Wetting behaviour split the coatings cleanly. Diatom, AgNP and AgNP-diatom films spread the droplet immediately, too fast for a contact angle to be measured at all. Glass measured 24° and the egg albumin film 62.9°, placing the protein film in a different wetting regime from everything else on the slide.

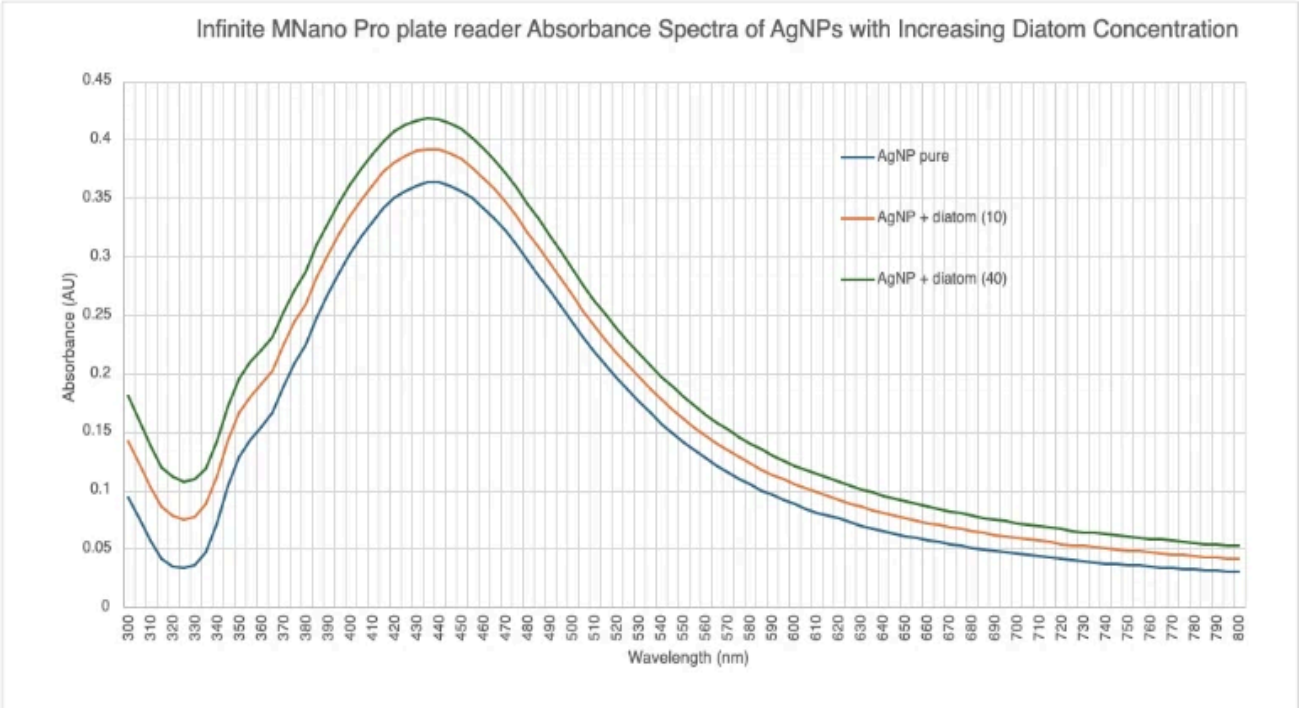


Figure 4.1 Absorbance of the silver nanoparticles at increasing diatom concentration. The plasmon peak holds at 435 nm and rises from 0.363 AU to 0.422 AU, so the diatoms change how much light scatters without shifting where the particles absorb.

WHERE THE HYPOTHESES FAILED

The adhesive. Egg albumin is reported to reach ultrahigh adhesive strength and to outperform commercial glue on glass and tissue. On leather it yielded at 0.237 MPa against 9.8 MPa for commercial glue, with a modulus of 0.0158 MPa against 0.2989 MPa, roughly forty times weaker. Two explanations survive. Substrate dependence, since the published comparison was made on glass and tissue rather than leather. And unoptimized preparation, since curing time and rehydration ratio were fixed rather than tuned. Both remain open, so the result is a rejection of the hypothesis rather than a refutation of the published work.

The electrode. Silver-modified carbon electrodes should raise peak current and narrow peak separation, both signs of faster electron transfer. Measured on a three-electrode ferricyanide cell against Ag/AgCl, silver loading raised peak current from 0.0027 mA to 0.0028 mA, a 3.7% change, while peak separation widened from 0.28 V to 0.32 V. Reversibility got worse, the opposite of the published result. The bright-field images give the reason. Deposition was visibly non-uniform, thick dark regions beside thin ones, so the working surface was never the uniform film the comparison assumes. The finding is a deposition problem rather than a materials one, and the fix is a thinner, more uniform coating over a conductive underlayer.

MISSING CONTROL

The absorbance series has no diatom-only blank, so the rise in peak absorbance with loading cannot be fully separated from scattering by the diatoms themselves. That control is the first thing to add before the enhancement claim is made quantitatively.

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Polymer Power: Crosslinker Concentration and PDMS Mechanical Tunability

The hypothesis failed, and the instrument was tested before the data were interpreted. It deviated from literature by 2% to 587%, which is the finding that governs the dataset.

PDMS stiffness is set by crosslink density, and crosslink density is set by the base to curing agent ratio, which is exactly why the material gets used across microfluidics and organ-on-chip work. It is tunable. Samples at 20:1, 10:1 and 5:1 were cast, cured overnight at 65 °C and tested in tension and compression, with alginate-gelatin hydrogels and a set of engineering polymers and metals alongside them. The hypothesis was the standard one: more crosslinker, higher modulus.

VALIDATING THE APPARATUS FIRST

Before any PDMS number was used, the Universal Mechanical Testing Kit was run against materials with known moduli.

Material	Measured modulus	Literature	Deviation
Polystyrene (bend)	3294 ± 260 MPa	3370 MPa	2%
PETG (bend)	8705 ± 203 MPa	8900 MPa	2%
PLA (bend)	2936 ± 48 MPa	2060 to 2150 MPa	40%
PVC (bend)	16412 ± 1085 MPa	2390 MPa	587%
Polypropylene (tensile)	818 ± 58 MPa	1970 MPa	59%
PCL (tensile)	722 ± 105 MPa	340 MPa	112%
Aluminium (tensile)	234 ± 8 MPa	70,600 MPa	99.7%

Two materials came in at 2% and the rest scattered from 40% to 587%, with no consistent direction to the error. That rules out a calibration offset and points at the fixture. Soft samples slipped in the grips or failed inside the clamped region rather than in the gauge section, and a bend fixture that reads polystyrene correctly and PVC seven times high is not measuring what it claims to measure.

WHAT THE PDMS DATA SHOWED

Base to crosslinker	Tensile modulus	Compressive modulus	Compressive yield
20:1 (least crosslinker)	2.80 ± 0.68 MPa	2.58 ± 0.46 MPa	4.70 ± 1.80 MPa
10:1	1.24 ± 0.15 MPa	1.27 ± 0.18 MPa	2.33 ± 0.52 MPa
5:1 (most crosslinker)	1.25 ± 0.18 MPa	1.35 ± 0.55 MPa	5.72 ± 1.39 MPa

The lowest crosslinker density gave the highest modulus in both test modes, which inverts the prediction. Compression curves were the cleaner of the two, since flat platens give uniform contact and remove the gripping problem entirely.

TWO EXPLANATIONS, KEPT SEPARATE

The first is the instrument, and it accounts for most of the deviation. Published measurements on the same ratios, taken on a proper universal testing machine and by optical methods, disagree with these numbers and agree with each other.

The second is physical, and it would be interesting if it survived. Published hardness data show stiffness rising from 10:1 to 10:2 and then falling at 10:3, which is consistent with a critical crosslinker concentration above which excess curing agent stops acting as crosslinker and starts acting as base elastomer. A related study reports stress softening under the same conditions. Mixing and degassing inhomogeneities would push in the same direction, by lowering effective crosslink density in nominally high-crosslinker samples.

RECOMMENDATION

The PDMS data should not be used for tuning a biomaterial until the tests are repeated on a calibrated universal testing machine. Until the instrument is exonerated, the critical concentration hypothesis cannot be separated from apparatus error, and the report says so rather than choosing the more interesting of the two.

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Matrix Mastery: Optimizing Release Kinetics in Alginate Beads

Full factorial optimization more than doubled release time, 480 s to 1080 s, and the confirmation run landed inside the predicted interval at 92% mean circularity.

Problem

Alginate encapsulation is governed by several interacting process variables, and formulation decisions commonly get made one factor at a time. That cannot detect interactions and it cannot identify an optimum.

Approach

A 2^3 full factorial design over alginate concentration, surfactant and extrusion geometry, measured on four responses, optimized in a prediction profiler, then verified by a full release-kinetics study on the selected formulation against three competing transport models.

Result

Release time spanned 480 s to 1080 s across the design space. The predicted optimum was confirmed experimentally at 92% mean bead circularity, and release was identified as Higuchi-type Fickian diffusion for both payloads, at R^2 = 0.968 and 0.949 against 0.91 and 0.83 for zero order.

THE DESIGN MATRIX

Factor	Low level	High level
Sodium alginate	1% w/v	3% w/v
Tween 80 surfactant	Absent	Present
Extrusion geometry	18-gauge tip, ≈0.8 mm	Untipped 3 mL syringe, ≈2 mm

Four responses were measured across all eight runs: release time, dissolved absorbance, mean circularity, and uniformity defined as the standard deviation of circularity over fifteen beads per treatment, imaged on a ChemiDoc gel imager against a ruler and measured in ImageJ.

Release time ranged from 480 s to 1080 s across the eight runs. That is a 2.25-times spread out of three binary factors. The JMP prediction profiler identified 3% alginate with surfactant through the 3 mL syringe as the slow-release optimum. It predicted 949 s, with a 701 to 1197 s interval, at 0.893 circularity.

VERIFICATION

The optimized formulation was manufactured and characterized against the prediction. Beads reached 0.920 mean circularity for the tartrazine payload and 0.903 for erythrosine, with uniformity of 0.032 and 0.015. Inside the predicted interval. Loading capacity was determined spectrophotometrically from a standard curve and worked through to 23.9% and 23.1% respectively.

Release was then run in 40 mL of PBS at 37 °C under 200 rpm agitation, sampled at intervals across 77 minutes, and fitted against zero-order, first-order and Higuchi models. Higuchi won for both payloads, which identifies the mechanism as Fickian diffusion through the matrix rather than surface erosion or first-order dissolution. The low intercept says burst release is minimal, and that is the property the formulation was optimized for.

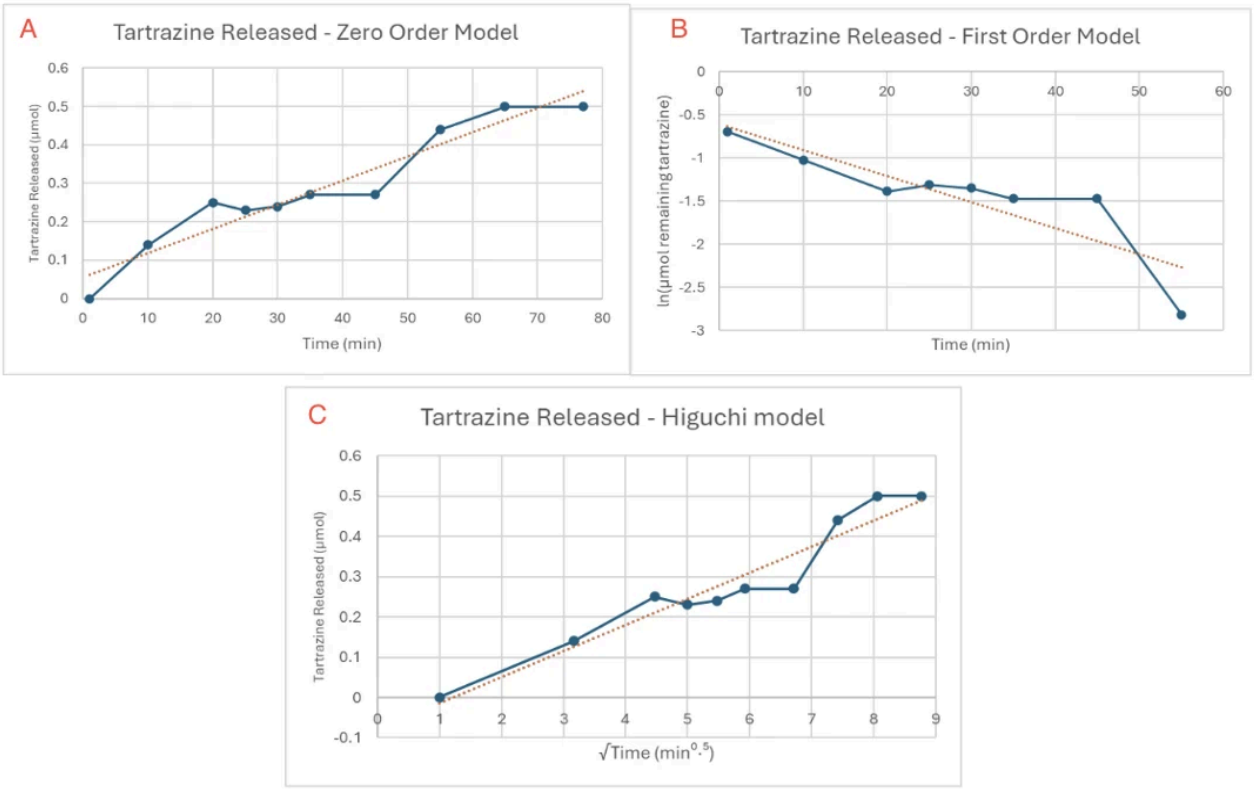


Figure 6.1 Tartrazine release fitted to zero-order, first-order and Higuchi models. Higuchi is the only one the data track across the full 77 minutes, which is what identifies the transport mechanism rather than just describing the curve.

THE RESULT THAT WOULD DRIVE THE NEXT DESIGN

The two dyes were expected to differ modestly. They did not. Tartrazine released 54% of its payload at 45 minutes against erythrosine's 100%, where roughly 40% had been predicted for erythrosine. Molecular size alone did not predict release rate. The larger molecule left the matrix faster, which points at interaction with the alginate network rather than steric hindrance as the governing factor, compounded by the different dissolution chemistries used for each dye.

LIMITS AND NEXT ITERATION

Temperature control drifted ±2 °C around 37 °C. Extrusion was manual, so droplet size carried operator variance. Erythrosine fluorescence had to be restricted to a validated linear range because of inner-filter effects above roughly 20 µM. And the Higuchi model is known to fit poorly where burst release is present, so its win here is partly a consequence of the formulation selected.

A syringe pump for extrusion, a jacketed vessel to hold temperature, and both dyes run in a single dissolution medium to separate the payload effect from the chemistry confound.

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Calibrating for Confidence: Ensuring Precision in Analytical Measurements

Instruments traced to reference standards, and two systematic effects quantified rather than absorbed into the result. Balances tracked to $R^2 = 0.9999$.

Every number produced downstream in a lab depends on instruments nobody rechecks. This module established the traceability first, then used the calibrated instruments to build absorbance and fluorescence curves and solve for unknown concentrations.

INSTRUMENTS AGAINST REFERENCE STANDARDS

Instrument	Range checked	Outcome
Analytical balance	2 to 50 g, triplicate	$R^2 = 0.9999$; largest deviation 0.0014 g at 2 g, or 0.07%
Precision balance	20 to 120 g, triplicate	$R^2 \approx 0.999$; overestimates by 0.13 g at 60 g and 0.27 g at 80 g, all under 0.3%
200 µL micropipette	Six volumes, triplicate, gravimetric	Under 1% deviation on water, within manufacturer tolerance
pH meter	Three-point at pH 4, 7, 10	Maximum error 0.03%, well inside the ±0.05 unit accuracy expected

The precision balance is the useful comparison. It is accurate enough for its purpose and measurably less repeatable than the analytical balance at the low end of its range, which is the difference between an instrument being wrong and an instrument being used outside where it performs best. A 10× PBS stock prepared with these instruments came out with every salt mass within 0.5% of target and a final pH of 7.408 against a 7.40 target.

TWO SYSTEMATIC EFFECTS, ISOLATED AND QUANTIFIED

Viscosity. The same pipette that delivers water to better than 1% delivered 50% glycerol at 103.0 µL against a 100 µL target. That is a consistent 3% overdelivery, from resistance during aspiration and adhesion inside the tip. It is a bias and not scatter, so it propagates into every solution made with a viscous reagent unless the technique changes.

The inner-filter effect. Erythrosine fluorescence rose linearly to about 20 µM and then flattened, because at high concentration the sample absorbs both excitation and emission light before either reaches the detector. Reading an unknown off the flat portion of that curve would return a wildly wrong concentration with no visible sign that anything was wrong. The unknown was therefore solved inside the validated linear range instead, at $R^2 = 0.9698$, giving 40.9 µM.

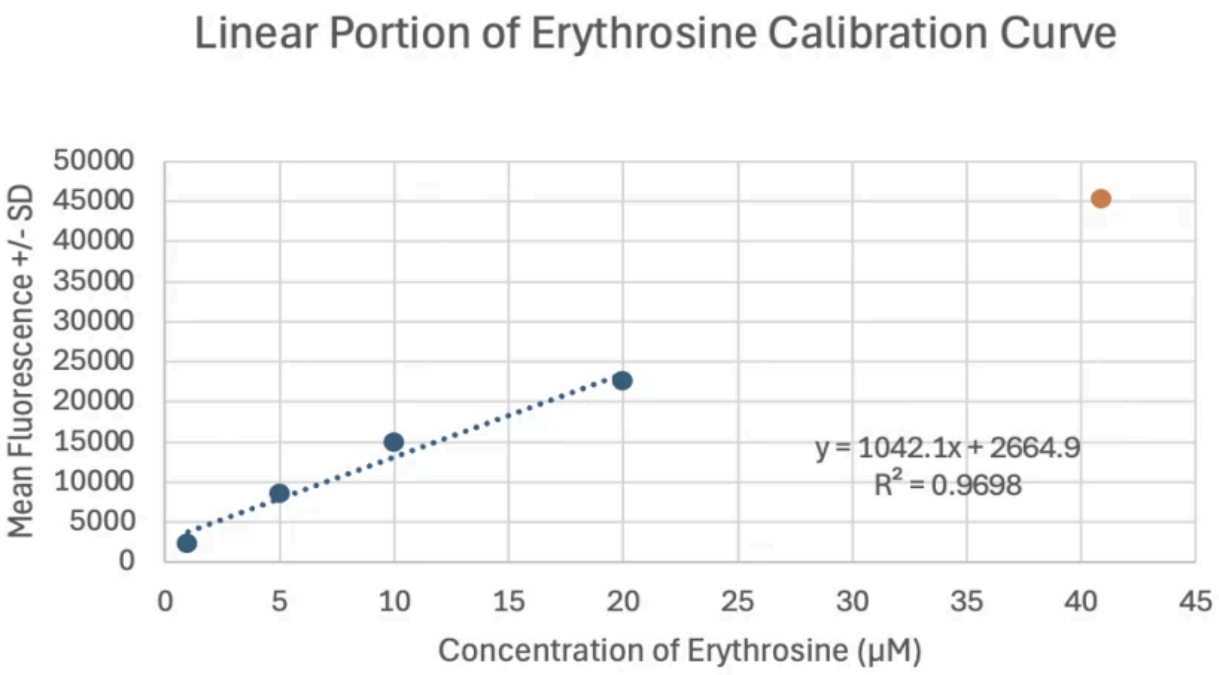


Figure 7.1 The linear portion of the erythrosine calibration curve, fitted at $R^2 = 0.9698$, with the unknown solved from the fit at 40.9 µM. Everything above roughly 20 µM was excluded before fitting, not after.

ABSORBANCE, AND A DISCREPANCY LEFT STANDING

Tartrazine absorbance against concentration followed Beer-Lambert at $R^2 = 0.982$ and gave the unknown at 60.7 µM. The peak wavelength, though, came out at 370 nm where the literature reports roughly 425 to 430 nm. The linear relationship at that peak is sound, so the calibration curve stands. The wavelength discrepancy is reported rather than explained away: plate scattering, a scan or dilution error and contamination are all consistent with it, and distinguishing between them requires a finer wavelength scan than this module ran.

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Tipping the Needle: A Dissolving Microneedle Patch to Reduce Injections

A cone pushed 80% of the way into skin leaves 48.8% of its volume outside it. Putting the drug only in the tip is what turns a variable dose into one that can be specified.

Problem

Teriparatide rebuilds bone in osteoporosis, but it is given as a daily injection for up to two years. In one claims analysis only 43.1% of patients were still taking it after a year, and fear of needles is a documented reason people avoid injectable treatment.

Approach

A dissolving microneedle patch that holds the drug in a water-soluble polymer and releases it as the needles dissolve in the skin. Needle dimensions, spacing and drug placement were chosen from published delivery data and design calculations, then set against four evaluation criteria.

Working thesis

A 14 × 14 array of conical needles, 700 µm tall on a 250 µm base at 600 µm spacing, with the drug confined to the distal four-fifths of each needle, so that the drug loaded is the drug inserted.

THE TREATMENT BURDEN

Osteoporosis weakens bone until it fractures under ordinary loads, and an estimated 9 million osteoporotic fractures occurred worldwide in 2000 alone. Teriparatide is a treatment that builds bone rather than only slowing its loss. The cost is a daily injection that can continue for two years, more than 700 injections in a single course.

Persistence is where it fails. In a Japanese claims database analysis, only 43.1% of patients were still on daily teriparatide past one year. Needle fear is one documented reason. In an international survey of more than 2,000 adults, 63.2% reported some level of fear of needles, and 18.3% of those had avoided an injection for a serious medical condition. The drug works when it is taken. The delivery route is what patients drop.

WHAT THE PATCH LITERATURE ALREADY SHOWS

A microneedle patch carries medication through the outer skin barrier on projections a fraction of a millimetre long. Human studies have already delivered teriparatide this way and measured increases in bone mineral density, so the open question is not whether the route works. It is whether it delivers enough, consistently.

The clearest warning comes from a related drug. In the wearABLe trial, a patch carrying 300 µg of abaloparatide raised lumbar spine bone mineral density by 7.14% after 12 months, against 10.86% for an 80 µg injection. The patch held 3.75 times more drug and produced less bone. Drug loaded onto a patch is not drug delivered, and closing that gap is the design problem.

WHERE THE DOSE IS LOST

A cone's volume scales with the cube of its height measured from the tip. A needle inserted to 80% of its height therefore places only 0.8^3, or 51.2%, of its volume below the skin surface, and leaves 48.8% outside it. Under uniform loading, nearly half the drug never crosses the barrier.

The second cost is variability. Skin thickness fluctuates at the micron scale, so under uniform loading the delivered fraction moves with every application. Confining the drug to the distal four-fifths of each needle removes both problems at once. With the drug only in the part that will be inserted, the dose stops depending on exactly how far each needle went, and the loading can be shifted further into the tip if testing shows it is needed. Two-layer dissolving needles of this kind, a drug-containing tip layer cast separately from the backing, have already been fabricated for insulin.

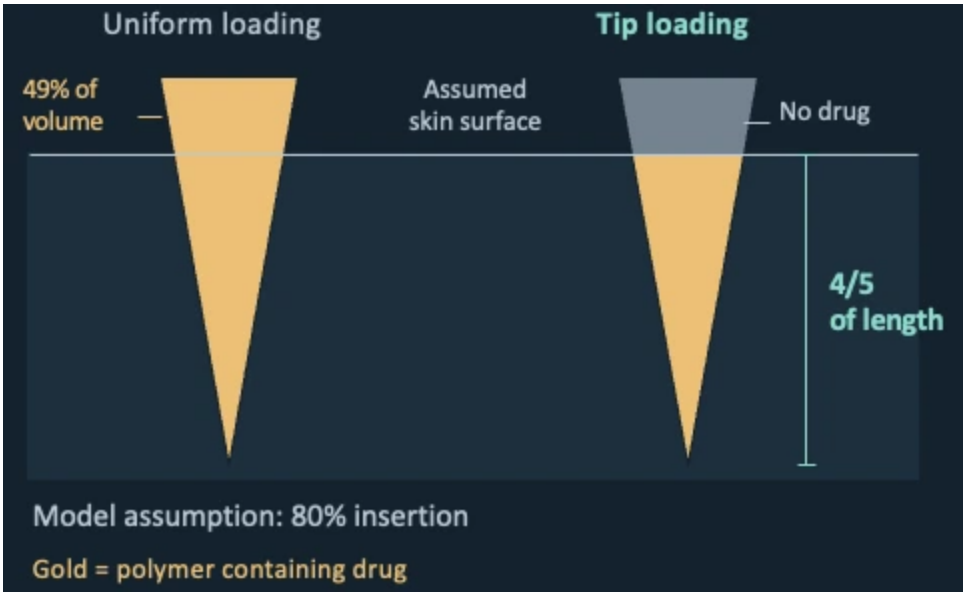


Figure 8.1 Uniform loading against tip loading, under the model assumption of 80% insertion. A uniformly loaded cone leaves 49% of its drug-bearing volume above the skin. Confining the drug to the distal four-fifths of the needle places all of it below the surface. Gold is polymer containing drug.

THE PROPOSED ARRAY

The starting design is a dime-sized patch carrying a 14 × 14 array of conical needles, 196 in total, each 700 µm tall on a 250 µm base diameter, with centres 600 µm apart. The geometry is checked against a published review of optimal dissolving microneedle geometry rather than chosen in isolation.

Every dimension trades against another. Larger needles or more of them hold more drug, but both make complete insertion harder, which raises the application force or calls for an applicator. A 20 N application force shared evenly across 196 needles gives roughly 0.1 N per needle, and whether that inserts each one fully without breaking it is the first thing the analysis has to establish.

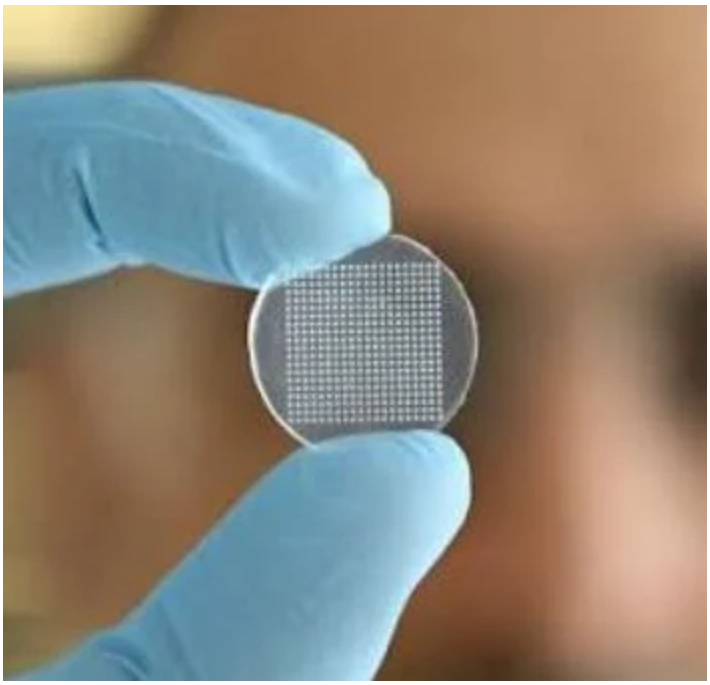


Figure 8.2 A dissolving microneedle patch held for scale, representative of the format proposed rather than the proposed design itself.

FOUR CRITERIA THE DESIGN HAS TO PASS

Each criterion targets a specific way the patch could fail once it leaves the lab.

Criterion	What is compared	Target
Delivered dose and consistency	Active drug reaching the bloodstream, and its variation from one application to the next	Reliable exposure comparable to a 20 µg injection
Strength margin	Force that breaks a needle, against the force needed to insert it	A wide gap, for a reliable application
Application force	Force the patch requires, against the thumb force of the least capable patients	Within about 20 N, or an applicator
Drug stability	Teriparatide activity after fabrication and storage, including temperature fluctuation	Activity retained in the patch

WHAT THE PROPOSAL CLAIMS, AND WHAT IT DOES NOT

This is a written research proposal, not a fabricated or tested patch. The array dimensions are a starting point for testing rather than an optimum, the 80% insertion depth is a model assumption, and 0.1 N per needle assumes the load is shared evenly across the array. Testing would adjust needle height, loading depth and array density against the four criteria.

Sterilization, dose consistency and infection risk all need evaluating for this patch specifically. The case for home use is supported by a phase 1 trial in which adults applied a microneedle influenza vaccine patch themselves. If the design holds, the same architecture is a model for delivering other peptides and medications that currently need an injection.

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09 · METHOD DEVELOPMENT · APRIL 2026 · THREE-PERSON TEAM

In-Situ Substrate Stiffness Calibration for Traction Force Microscopy Using Magnetic Beads

Replacing an assumed substrate stiffness with one measured during the experiment. The whole error budget collapses to one relation, and that bounds modulus uncertainty at 1 to 10%.

Problem

Traction force microscopy converts bead displacements into cell forces by inverting an elastic model that needs the substrate's Young's modulus. In practice that modulus gets measured once, in bulk. Hydrogels vary locally with crosslinking, swelling, aging and temperature, and cells probe local zones, so the mismatch enters every reconstructed force as systematic bias rather than as noise.

Approach

A sparse second population of magnetic beads is embedded beneath the cell contact surface, actuated between measurements with a known magnetic force, and the resulting displacement imaged with the optics already trained on the sample. The substrate reports its own local stiffness during the experiment.

Result

An error analysis that reduces the method's accuracy to a single relation, where relative modulus error equals relative displacement error. That converts directly into a design rule and bounds achievable modulus uncertainty at 1 to 10%.

WHERE THE PROBLEM CAME FROM

This came out of a critical review of the foundational traction force microscopy paper, which measured collective cell migration on a 1.3 kPa polyacrylamide gel and concluded that migration is driven by a global tensile tug of war rather than by leader cells. That review argued the paper's substrate assumptions bounded what its data could support. A purely elastic gel standing in for viscoelastic tissue, a 2D inversion discarding out-of-plane deformation, and a single bulk modulus applied across a heterogeneous surface. This project addresses the third of those.

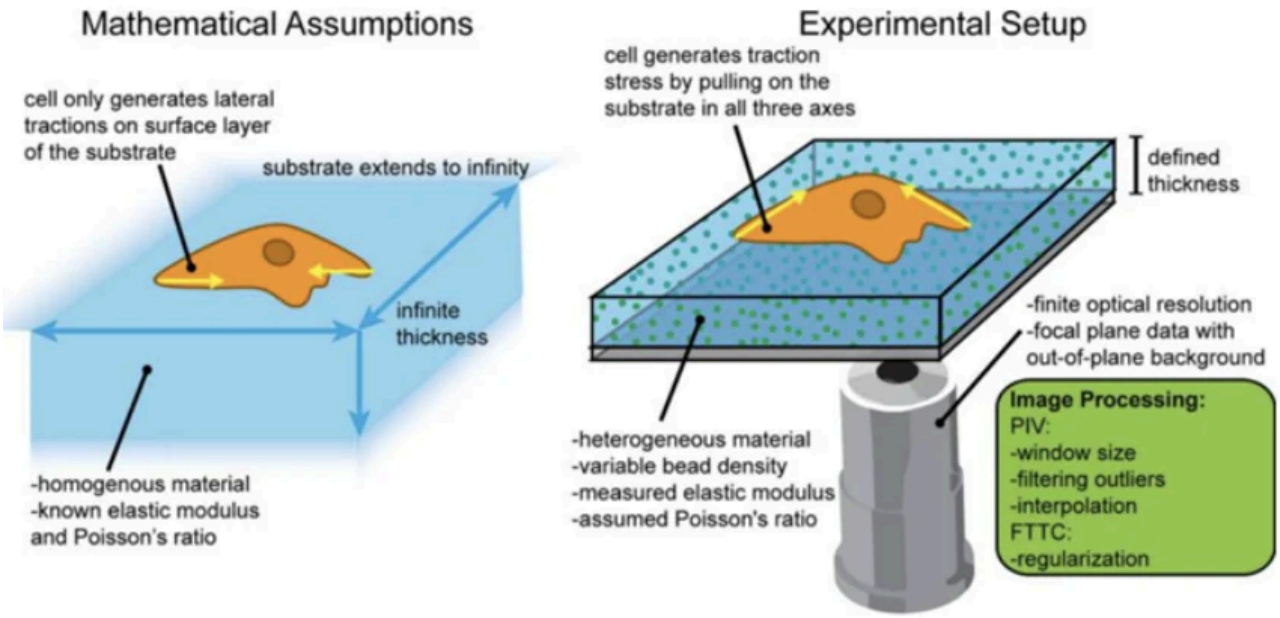


Figure 9.1 The gap the method addresses. The inversion assumes a homogeneous half-space of known modulus and Poisson's ratio. The experiment supplies a finite, heterogeneous gel whose modulus was measured once, in bulk.

WHY THE INPUT MATTERS SO MUCH

The forward problem is a convolution of traction against the substrate's Green's function, discretized to a linear system whose direct inversion is unstable because the matrix is ill-conditioned. Reconstruction therefore uses Tikhonov regularization. The regularization parameter trades noise amplification against underestimated traction peaks, which makes the choice one of which error to carry. An inversion that already amplifies whatever error it is given is the worst possible place to feed an assumed constant.

THE CALIBRATION CHAIN

Three relations connect the applied field to a local modulus. Force on a paramagnetic bead scales with bead volume, magnetic susceptibility, and the product of field and field gradient, so it is tunable through electromagnet current or magnet distance. The elastic solution for a rigid sphere embedded in an infinite medium converts that force and the measured displacement into a local shear modulus. Assuming incompressibility, Young's modulus follows as three times the shear modulus.

THE ERROR BUDGET COLLAPSES TO ONE TERM

Because modulus scales inversely with displacement, differentiating the relation collapses the entire error budget to $\Delta E/E = \Delta u/u$. Relative modulus error equals relative displacement error, and nothing else. That is immediately actionable. At a typical positional accuracy of about 10 nm, a 100 nm induced displacement carries roughly 10% modulus uncertainty, while pushing displacement to 1 µm on the same instrument drops it to 1%.

The same relation bounds where the method applies. Stiff substrates displace too little to resolve above the noise floor, and very soft ones displace far enough to leave the linear elastic regime. Soft regions therefore measure more precisely than stiff ones. That is a real constraint, and one the technique has to state up front rather than discover later.

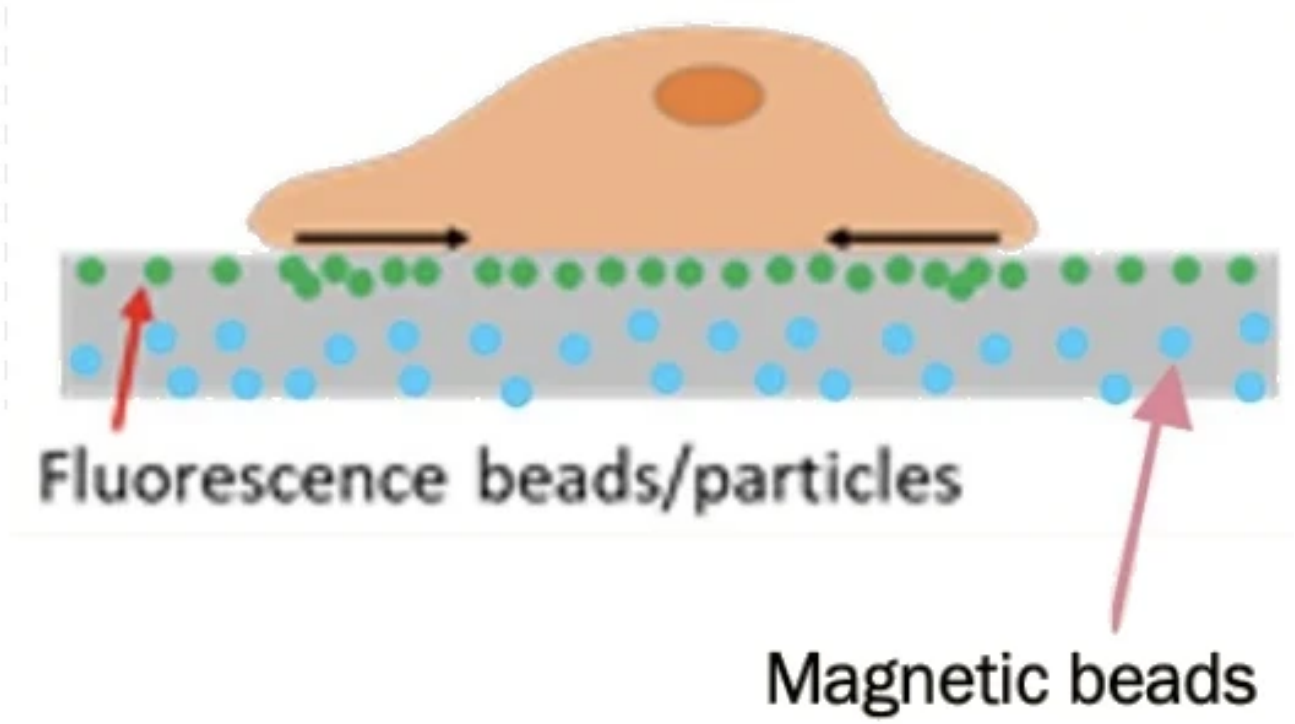


Figure 9.2 The two bead populations. Fluorescent beads at the contact surface report cell traction; magnetic beads below it report local substrate stiffness.

WORKING AROUND THE CELL

Magnetic beads cannot be actuated during traction measurement, because that corrupts the displacement field being read. Calibration runs afterwards as brief, reversible compliance checkpoints, kept small and infrequent enough not to mechanically stimulate the cell. The magnetic population stays sparse, since it only needs to probe local zones rather than resolve a traction field.

WHAT THE METHOD CLAIMS, AND WHAT IT DOES NOT

Actuation force varies bead to bead with orientation, which trades one spatial uncertainty for another. Real gels are viscoelastic, so a pulse-and-measure model neglects creep and delayed relaxation. The calibration beads carry their own uncertainty in size, roundness and susceptibility.

The claim is not that the beads remove uncertainty. It is that they replace an assumed modulus with a measured estimate, which is a strictly better input to an inversion that amplifies whatever error it is given. Validation requires a gel of independently characterized heterogeneity, so the recovered stiffness map can be checked against ground truth.

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Reduced-Alphabet Redesign of a TIM Barrel: Twenty Amino Acids to Five

55% of a 247-residue sequence changed and 75% of the alphabet removed, and the core still aligned to 1 Å. The deviation went entirely into the loops.

Question

Modern proteins use twenty amino acids, but early proteins are thought to have used far fewer. How much of that alphabet a functional fold actually needs is still open, and it has direct consequences for de novo design.

Approach

The TIM barrel, eight parallel β-strands wrapped by eight α-helices and one of the most common folds in biology, redesigned twice from its 247-residue wild type: once with fifteen amino acids, then with five. Every substitution was justified residue by residue, and each design was validated by structure prediction against the wild-type baseline.

Result

The five-letter design changed 55% of the sequence while removing 75% of the alphabet, and still folded to the correct barrel topology with no strand-order swaps. Core secondary structure aligned to 1 Å, and the deviation concentrated entirely in the loops at 3 to 5 Å.

ESTABLISHING THE BASELINE FIRST

Structure prediction on the wild-type sequence reproduced the reference to approximately 1 Å RMSD with core confidence in the 90s. That mattered before anything was changed, because it establishes that deviations in the redesigns can be attributed to the redesign rather than to prediction error. Structural analysis then set the constraints any redesign had to respect: alternating burial along the β-strands, the catalytic pocket exception where two polar residues sit buried as a hydrogen-bonded pair, and the division between bottom loops that lock the structure and top loops that gate the active site.

FIFTEEN LETTERS

Five residues were removed, each substitution chosen to preserve a specific property while dropping a specific liability. Cysteine to serine, for identical geometry without the reactive sulfur. Methionine to leucine, for comparable hydrophobic volume without oxidation chemistry. Tryptophan to phenylalanine, for retained π-stacking at reduced bulk. Glutamine to asparagine, for an identical amide one carbon shorter. Arginine to lysine, keeping the positive charge at the cost of directional hydrogen bonding. Glycine and proline were kept as irreplaceable, and the catalytic triad was preserved outright.

The result diverges 11% from wild type and reproduced the fold at 1 to 2 Å backbone RMSD with core confidence of 90 to 100. The informative deviations were local and predictable. At the core, the smaller phenylalanine left a gap that a neighbouring side chain closed by shifting 0.5 Å. At the surface, replacing an arginine that had stabilized a loop through multiple hydrogen bonds left that loop measurably more flexible.

FIVE LETTERS

Alanine, glycine, isoleucine, glutamate, lysine. One generic hydrophobe, one turn residue, one small filler, one negative and one positive charge. The mapping rules were explicit and conditional on structure. Hydrophobics default to isoleucine unless surface exposed. Polar residues become glutamate when exposed and alanine when buried, because burying a charge costs more than losing a hydrogen bond. Charges map so as to preserve existing salt-bridge pairs.

The design decision worth defending is the loss of proline. With no rigid residue available, turn rigidity had to be engineered back in, so lysine and glutamate pairs were placed flanking flexible turns to substitute salt-bridge constraint for proline's ring geometry. The prediction shows exactly how far that compensation goes and where it stops. Overall Cα RMSD of 2 to 3 Å, but core secondary structure aligning to 1 Å with the entire deviation sitting in the loops at 3 to 5 Å. Helices came out longer and frayed at the ends, and the worst loop dropped below 50 confidence. That is a direct readout of the missing proline at the precise position where it used to sit.

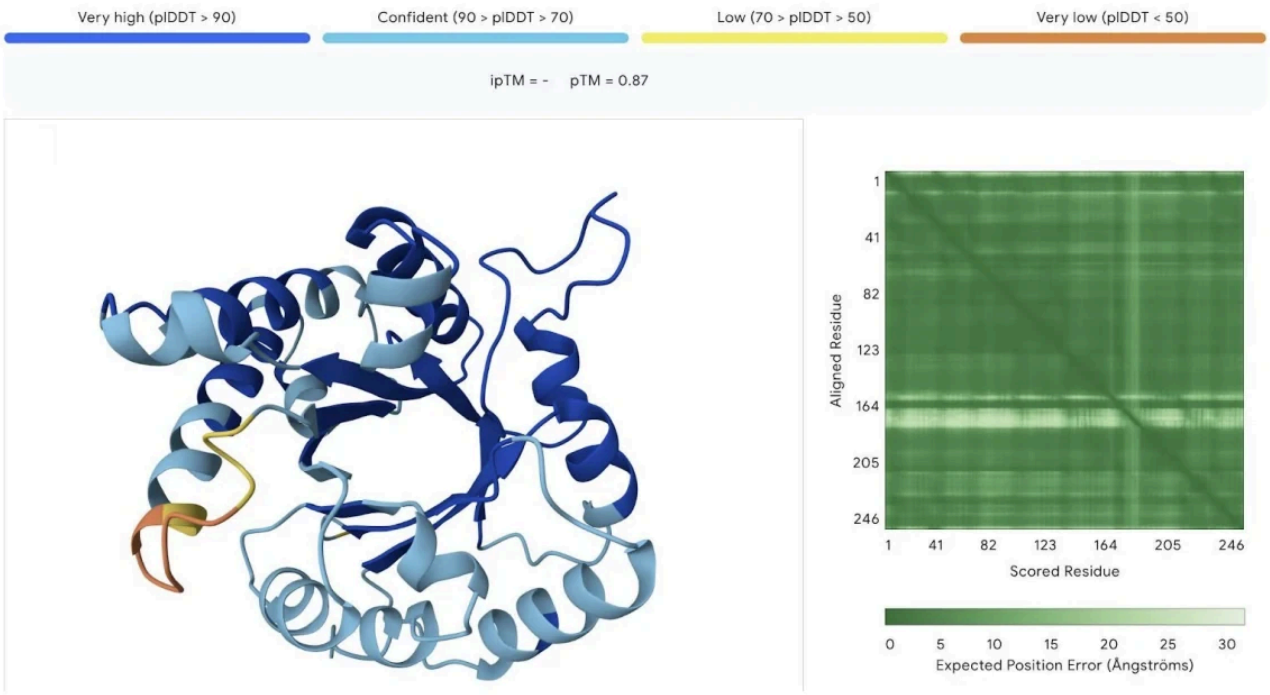


Figure 10.1 The redesigned barrel coloured by per-residue confidence, with its predicted aligned error matrix. Barrel topology is recovered at high confidence, and the low confidence sits in a surface loop rather than anywhere structural.

WHY FIVE IS THE FLOOR

Each further removal breaks a specific requirement. Without glycine, tight turns and helix-to-strand connectivity fail. Without isoleucine, the core underpacks because alanine is too small. Without either charge, balanced salt bridges and solubility are lost. Without alanine, charges get forced into the buried core.

LIMITS, AND THE TWO THINGS TO TEST FIRST

These are predicted structures, not solved ones, and prediction confidence is not thermodynamic stability. The design's stability, folding cooperativity and catalytic activity are all untested.

Two consequences follow directly from the sequence, and they are the first things to check experimentally. A surface composed almost entirely of glutamate and lysine narrows the pH window in which the protein stays soluble, at both ends. And a protein built from only five residue types would resist crystallography for lack of distinguishable species, which makes NMR the appropriate structural method precisely because there is so little to distinguish.

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A Bilayer Magnesium-Reinforced Bioglass Implant Coating Specified to Cortical Bone

A scaffold specified to 180 MPa that degrades into the cortical bone range over six weeks, at roughly $278 in materials against an $8,000 to $13,000 implant.

Problem

Titanium hip implants fail through three coupled mechanisms: metal ion release from wear driving inflammation and osteolysis, post-operative infection, and the revision surgery that follows both. Elderly patients tolerate that revision poorly.

Approach

A bilayer coating designed against the mechanical specification of human cortical bone. An inner magnesium rod scaffold set in bioglass cement carries load, beneath a mesoporous bioactive glass layer engineered for sustained, infection-triggered antibiotic release.

Result

A specification meeting the 100 to 200 MPa cortical bone compressive range, where every surveyed reinforcement strategy falls one to two orders of magnitude short, with a degradation schedule matched to bone regeneration and a materials cost thirty to fifty times below market implant price.

THE NUMBER EVERYTHING WAS DESIGNED AGAINST

Human cortical bone is 100 to 200 MPa in compression at 15 to 25 GPa elastic modulus. Every conventional route to reinforcing bioglass was checked against that figure and rejected on the evidence.

Reinforcement strategy	Why it was rejected
Sintered bioglass	An order of magnitude short, and requires 1250 °C processing
Polymer infiltration	Two orders short
Ceramic particle filling	Weakens tensile and flexural strength
Foam and porous scaffolds	Two orders short
Hydrogel composites	Four orders below 15 GPa
Metal particle reinforcement	Closest of the group, still outside specification

REINFORCED CONCRETE, AT IMPLANT SCALE

The load-bearing layer is magnesium rods playing rebar in a bioglass and magnesium phosphate cement. Bioglass cement on its own measures 16.0 MPa. With magnesium rods it reaches 180.0 MPa, then degrades to 100 MPa over six weeks. That puts it inside the cortical bone range at about the point regenerating bone resumes the load. Magnesium's corrosion is the mechanism here rather than the defect. Bare magnesium degrades in four weeks and PCL-coated magnesium in eight, against the four to six weeks bone needs to grow onto the implant.

The scaffold geometry is where the design moved past the literature. Rods rather than particles, since published work considered only particles and rods carry more load. Anisotropic pores, which reach 100 to 150 MPa where isotropic structures manage 2 to 12 MPa. And a hexagonal cell, which delivers 180 MPa under perpendicular compression at 50% porosity and carries the highest Weibull modulus of the geometries surveyed. That matters more than peak strength, because it means part-to-part consistency. Fatigue resistance comes out at 10^5 cycles at 3 to 30 MPa, against a physiological femoral load of 5 to 14 MPa.

ONE THICKNESS CONSTRAINT SETS THE WHOLE PROCESS ROUTE

Coating thickness had to land near 100 µm and stay under 200 µm total. Thinner degrades away, thicker delaminates. That single constraint fixed everything downstream. Thin magnesium wires. Viscosity reduction by shear thinning with polyethylene glycol 200. Bioglass ground to nanoparticles. Electrophoretic deposition with low-temperature annealing, because PCL melts at 60 °C while magnesium melts at 650 °C, so any high-temperature step destroys the layer beneath it.

A COATING THAT RESPONDS TO INFECTION WITHOUT SENSING IT

Conventional melt-processed bioglass releases its payload in a burst and then nothing, which is the wrong profile for infection prophylaxis. A mesoporous structure, 2 to 50 nm pores across a high surface area, loaded by soaking for 12 to 48 hours and applied by ten-minute electrophoretic deposition, releases gradually instead. The behaviour worth pointing at is that release is pH responsive by chemistry rather than by control logic. Infection acidifies the site, acid dissolves the coating faster, and antibiotic release rises. The device responds to infection without sensing anything at all.

COST

Roughly $100 of medical-grade titanium, $100 of bioglass, under $2 of magnesium, under $1 of PCL and about $75 of antibiotics. Approximately $278 in materials before manufacturing, against a market implant price of $8,000 to $13,000.

WHAT THIS IS, AND WHAT IT IS NOT

This is a specification supported by literature data, not a fabricated and tested article. Coating thinness feasibility, in vivo mechanical behaviour and cellular response are all unverified, and no study exists on a scaffold combining every component specified here.

In design-control terms, these are design inputs translated into design outputs, with literature evidence standing in for verification. Nothing here has been validated. The verification sequence would run simulated body fluid soaking, cell adhesion and proliferation assays, micro-CT for osseointegration, and full mechanical characterization under cyclic load.

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12 · GROUP DESIGN PROJECT · BIEN 220 · FALL 2024 · FIVE-PERSON TEAM

Bacterial Self-Healing Concrete: *Bacillus subtilis* Spores in pH-Triggered Microcapsules

The healed model displaced roughly six times less per unit load. The porosity that would have made the model realistic is the part that failed.

Concrete cracks, moisture and chloride ions get in, reinforcement corrodes, and the structure degrades. Epoxy injection and surface sealing are labour intensive, recurring and wasteful. A concrete that seals its own cracks removes the maintenance cycle instead of shortening it, which is the case for bioengineering a construction material.

THE HEALING MECHANISM, STEP BY STEP

Fresh concrete sits above pH 12, where *Bacillus subtilis* survives but works poorly, so the bacteria go in as spores. Dormant, formed under nutrient limitation at 37 °C over 48 to 72 hours, and far more durable than vegetative cells. The spores were encapsulated with urea and calcium nitrate in urea-formaldehyde microcapsules, synthesized by in-situ emulsion polymerization in a water-in-oil system at 800 to 1500 rpm and about 40 °C, then dried at 50 °C for 48 hours into 100 to 300 µm spheres and mixed in at 0.5% by weight, the loading reported to give uniform dispersion without compromising the matrix.

The trigger is the crack itself. Urea-formaldehyde is stable in alkaline conditions and degrades below pH 6. When a crack admits moisture and carbon dioxide, local pH falls from above 12 into the 7 to 9 range, the capsules break down, and spores and nutrients are released into the crack. Water and oxygen germinate the spores, urease-catalyzed hydrolysis of urea produces carbonate, carbonate meets the released calcium, and calcium carbonate precipitates in the gap. The material senses damage by chemistry alone. No sensing element, no external intervention.

THE MODEL

A 40 × 40 × 160 mm prism was modelled in Fusion 360 with a 1 mm through-crack, chosen because 970 µm is the widest crack bacterial healing has been shown to close. Two states were compared, cracked and 80% healed, 80% being the healing efficiency reported under moist curing. The face opposite the crack was fixed as a rigid body to represent a real support condition, and each state was loaded at 1,000, 3,000 and 5,000 N in two directions, front face aligned with the crack and side face.

Property	Aged concrete	Self-healing concrete
Young’s modulus	1200 MPa	1320 MPa
Yield strength	19.05 MPa	20.96 MPa
Ultimate tensile strength	4.00 MPa	6.52 MPa
Poisson’s ratio	0.15	0.20

Material properties were taken from published measurements on *B. subtilis* concrete rather than assumed, and the healed region was modelled as continuous with the matrix rather than as a separate material, on the assumption that precipitation restores structural continuity.

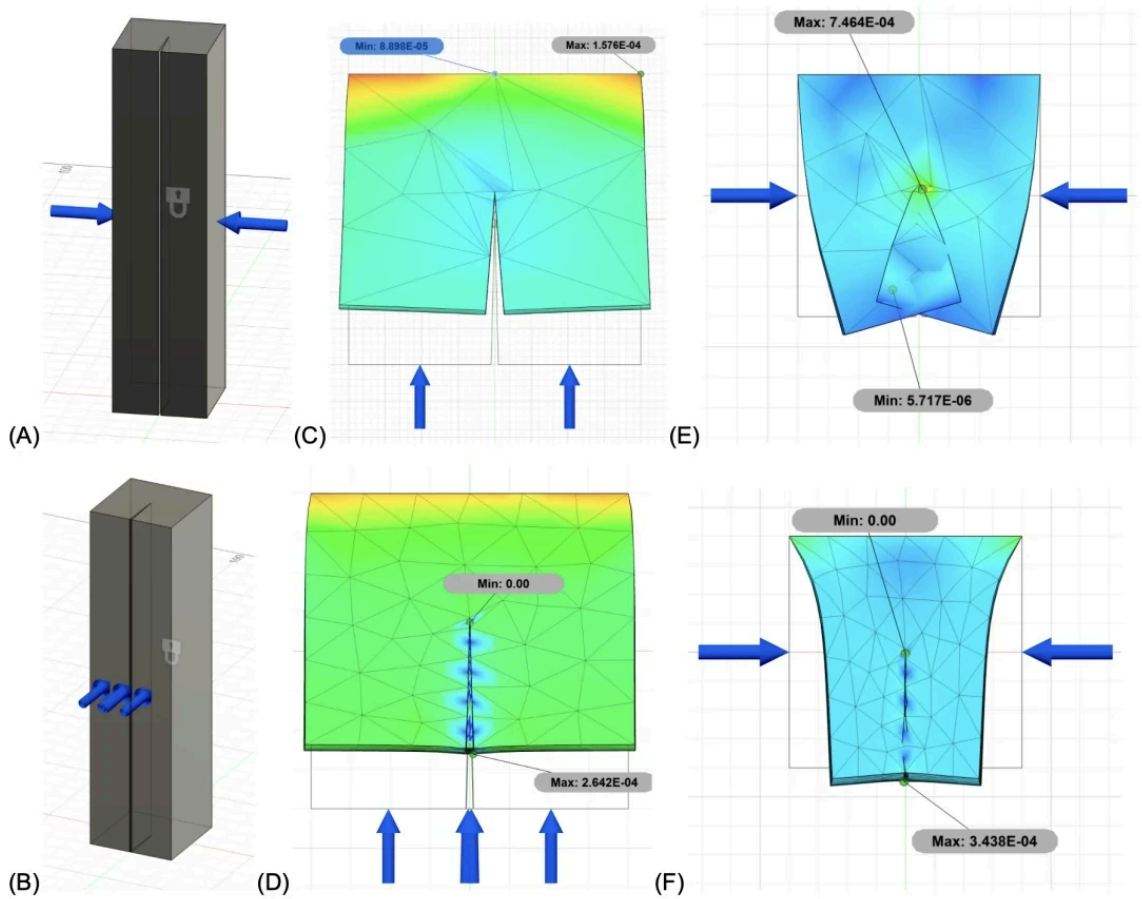


Figure 12.1 Fusion 360 static stress models. Side face and front face loading conditions, with deformation of the cracked prism against the 80% healed one under each. Panels A and B show side-face and front-face loading; C and D compare the cracked and healed prisms under front-face load, and E and F under side-face load.

RESULTS

Displacement is the clearest separator. Load against displacement gave a slope of 1.28×10^{-5} for the cracked prism and 2.00×10^{-6} for the healed one, a factor of roughly six, and even at 5,000 N the healed prism moved only 0.01 mm. Von Mises stress in the healed body ran 1.7 times lower, which says the internal load path is intact rather than concentrating at the crack tip.

The two loading directions behaved differently in a way worth noting. Under front face load, aligned with the crack, the cracked prism opened further while the healed geometry held its shape. Under side face load the cracked prism distorted substantially, and the healed one deformed moderately while keeping the crack region stable. Higher contact force in the healed model reads as better load transfer through a densified structure rather than as a penalty.

THE LIMITATION THAT MATTERS

Real concrete is not homogeneous. It is a matrix of pores and microcracks, and modelling that geometry was attempted and failed, because the pore structure was too complex for the simulation to resolve. The geometry had to be simplified to a homogeneous solid.

Capillary pores below 50 nm have little effect on compressive strength, so the simplification is defensible for this comparison. The model does not represent real concrete microstructure, however, and the results should be read as a comparison between two idealized states rather than as predicted field performance.

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Biosignal Analysis of Brain Activity and Cerebral Hemodynamics

Thirty-six connectivity matrices, computed to test whether the two-lobe structure visible in broadband EEG survives band decomposition. It does not.

A two-part study. Part one treats resting-state EEG as a spectral estimation problem. Part two treats the cerebral circulation as a system with an input and an output.

PART ONE: RESTING-STATE EEG

Four electrodes were selected to sample the scalp, frontal, central, occipital and left temporal, and 30-second segments compared between eyes open and eyes closed. Welch spectral estimation was benchmarked against the raw periodogram. The periodogram is jagged and high variance; Welch trades a little frequency resolution for a large reduction in variance by averaging overlapping windows. The alpha peak survives both. Segment length was varied at 10, 30 and 60 seconds, which shows resolution improving and the peak stabilizing as the window grows.

A five-band filter bank of second-order Butterworth filters was applied with zero-phase forward and backward filtering. Second order because the passband stays maximally flat without ripple, while higher orders risk numerical instability. Occipital alpha dominates eyes closed, and power shifts toward beta and gamma eyes open. Spectrograms on 10-second windows at 50% overlap show the pattern drifting across a 300-second recording, so the signal is treated as non-stationary overall and approximately stationary inside each window. A beta-to-alpha ratio was then tracked as a vigilance index over five minutes. It fluctuates continuously rather than holding a level, and runs higher eyes open, which is what visual engagement would predict.

THE EXTENSION

Broadband connectivity, computed as maximum absolute cross-correlation across every electrode pair, splits the scalp into two blocks. Frontal electrodes in one, parietal and occipital in the other, with central and frontocentral electrodes sitting between them as an intermediary. That structure holds steady across consecutive segments, which is what a resting brain should look like.

The question this raised, beyond what the assignment asked for, is whether that two-block structure is a real feature or an artifact of averaging everything together. Answering it meant computing roughly 1,700 electrode pairs per segment, over six consecutive segments, in each of six frequency bands. Thirty-six connectivity matrices.

It does not survive intact. Delta and theta run closer to the broadband picture and higher in overall correlation, and theta is the least differentiated of the six. Gamma shows the sharpest front-to-back separation, with almost no correlation across the divide. Alpha does something different again. It splits the posterior block along left and right hemispheric lines, so the division there is hemispheric and not parietal against occipital. The reading that fits is that higher-frequency bands carry more information per segment and therefore more discrimination between regions, so correlation falls exactly where processing is most differentiated.

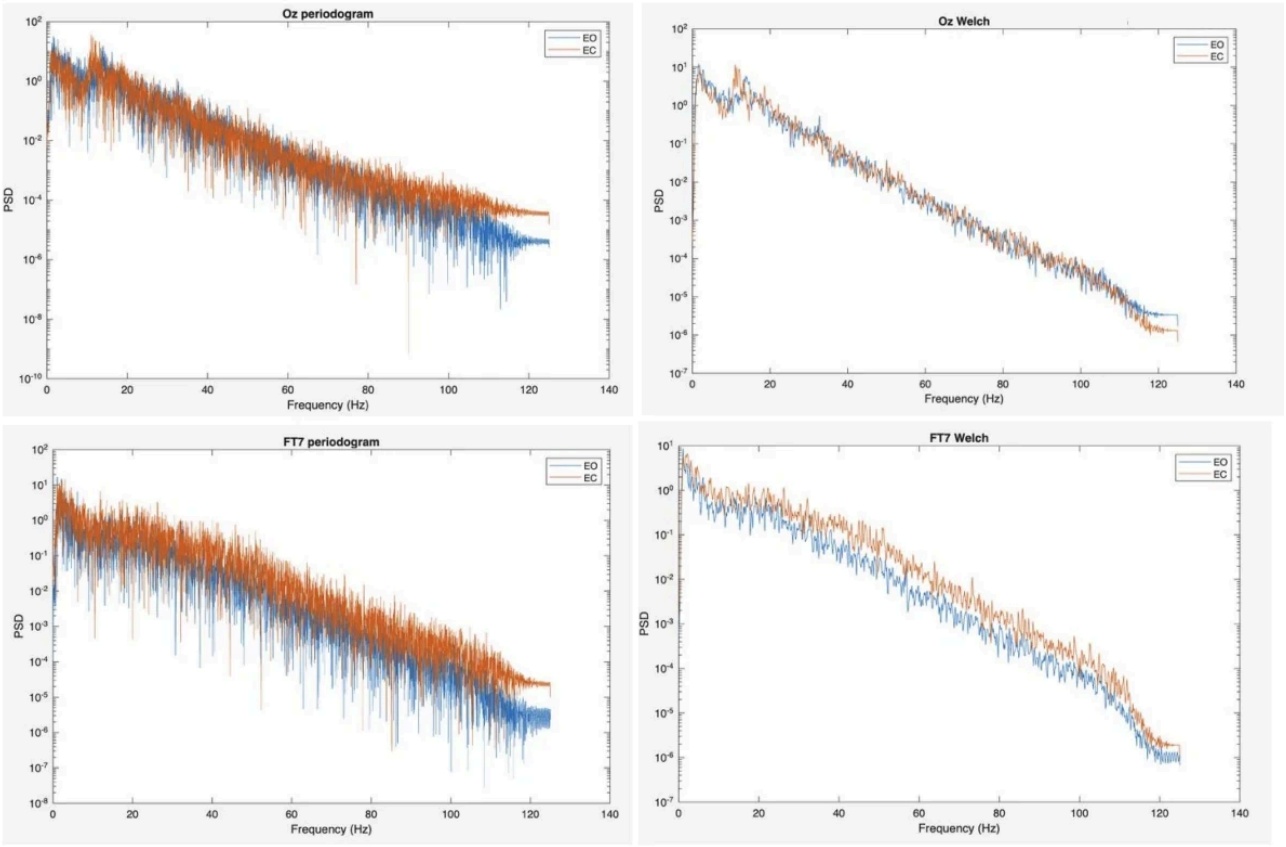


Figure 13.1 Occipital and left temporal power spectra, eyes open against eyes closed, estimated by raw periodogram and by Welch. Welch removes most of the variance and keeps the spectral features that matter.

PART TWO: PRESSURE IN, FLOW OUT

Arterial blood pressure was treated as the input and cerebral blood flow velocity as the output. Band-passed to the cardiac range at 0.5 to 10 Hz, the two correlate at 0.968 with flow lagging pressure by 0.065 s, the delay expected from wave propagation and vessel compliance. Both spectra peak at 1 to 1.5 Hz with harmonics above, and flow carries visibly less high-frequency power than pressure, since the cerebral vasculature damps fast pressure changes before they reach the flow signal.

On the beat-to-beat averaged data, where only slow fluctuations below 0.5 Hz remain, the two signals decouple at the lowest frequencies. Coherence is low below roughly 0.025 Hz, higher through the 0.025 to 0.3 Hz band, and the apparent rise above 0.4 Hz is estimation uncertainty rather than physiology. That low-frequency decoupling is autoregulation doing its job, with vessels adjusting diameter to hold flow steady while pressure moves.

A transfer-function model built by inverse Fourier transform reproduced slow trends and failed on rapid transients. That failure is the expected signature of fitting a time-varying, nonlinear autoregulatory system with a linear time-invariant model, and it is the more informative half of the result.

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Energy Footprint: A Seven-Day Energy and Carbon Audit Scaled to a Full Year

The largest energy flow in the audit, heating a Montreal house, emits almost nothing. Six drives home and three flights are 45% of the energy and 69% of the carbon.

Question

How much energy does one student's life take, where does it go, and how much of it reaches the atmosphere? The audit followed September 10 to 16, 2026, in term and in lacrosse season, living with four roommates in a downtown Montreal duplex, with summers in a family house in Longmeadow, Massachusetts.

Approach

A daily log of travel, activities and appliance use, with building heating and cooling derived from utility bills, heating degree days and McGill's campus energy inventory. The week was annualized, then converted to CO_2, embodied food energy, cost and human-labour equivalents.

Result

273 kWh for the week and 22,700 kWh a year once travel is included. Annual CO_2 of 3.76 t is dominated by travel, because Quebec's hydroelectric grid makes the biggest energy flow, home heating, almost carbon-free.

SCOPE AND ALLOCATION

Energy is reported as delivered energy, the kilowatt-hours that arrive at the meter, pump or tank, and annual figures are divided by 365 for daily values. Walking, the main way of getting around, is excluded. Allocation follows two rules. Personal use counts in full: showers, cooking, laundry, screens, and television even when roommates are watching, which makes it an upper bound. Continuous loads that run whether anyone is home, the refrigerator, water-tank standby and router, are split five ways in Montreal and six in Longmeadow.

Emissions use fuel factors from Canada's National Inventory Report: 2.307 kg of CO_2 per litre of gasoline, 2.681 for diesel, 2.560 for jet fuel and 1.926 per cubic metre of natural gas. Electricity is where location takes over. Quebec's 2023 consumption intensity is 1.9 g CO_2e per kWh, while the New England grid runs at 0.245 kg per kWh, roughly 130 times higher.

HEATING, DERIVED FROM A UTILITY BILL

Building heat is the largest flow and the one that cannot be read off a meter, so it was modelled. The duplex bill averages $400 a month, which Hydro-Québec's Rate D converts to 41,300 kWh a year. Removing water heating, appliance and lighting loads for five people leaves 26,540 kWh for space heating, and spreading that over Montreal's 4,363 heating degree days gives the house a heat loss coefficient of 253 W/K. The lease runs twelve months and the house is heated whether anyone is in it, so one fifth of the annual load is allocated, plus a personal window air conditioner.

Campus buildings were allocated per occupant-hour. McGill's 2024 inventory lists 15.3 million m^3 of natural gas, 182 GWh of electricity and 0.27 million litres of heating oil. The McConnell Engineering Building, nine storeys and about 16,000 m^2, uses 3.31 GWh a year, and an occupancy model of 2.27 million occupant-hours puts it at 1.46 kWh per occupant-hour, against roughly 645 hours a year of personal use. The Currie Gymnasium and Tomlinson Fieldhouse come to 2.72 kWh per occupant-hour, over 408 hours of training and treatment.

The Longmeadow house, 418 m^2 with electric resistance heat and central air, was modelled at an assumed 1.25 W/m^2K, or 522 W/K. Against 5,711 °F heating degree days that gives 39,800 kWh a year. A one-sixth share is assigned only for the 129 days spent there, which carry 21.2% of the annual heating load.

THE AUDIT WEEK

The week came to 273 kWh, or 39.0 kWh a day. Heating and cooling made up 66% at 181 kWh, appliances and electronics 18% at 49 kWh, and transport 16% at 43 kWh, all of it from one team bus trip to Ottawa for a lacrosse game. That trip, 382 km in a motorcoach burning 42.2 L per 100 km shared among 40 travellers, took Sunday to 77.1 kWh against 30 to 35 kWh on every other day.

Among appliances, water heating dominated at 22 kWh, 45% of the category, and it tracked the days with extra showers and laundry. Laptop and phone together drew 0.68 kWh for the whole week. Of the 9,420 kWh a year of heating and cooling, 57% came from the Montreal duplex, 22% from McGill buildings and 21% from the Massachusetts house.

ANNUALIZED, TRAVEL TAKES OVER

The audit week contained no car or air travel and entered building heat only as an annual average. Annualizing every category raises the total to 22,700 kWh a year, 62.2 kWh a day, 1.6 times the audit-week rate. The year splits three ways: the academic year in Montreal, on foot and heated by hydroelectricity; summer in Longmeadow, on a New England grid that runs mostly on natural gas; and the trips between them.

Source	Annual energy (kWh)	Annual CO_2 (kg)
Montreal duplex, heating and cooling	5,365	10
McGill buildings	2,048	339
Longmeadow house, heating and cooling	2,012	493
Appliances and electronics	2,549	223
Car, six round trips home	5,730	1,373
Three flights	4,569	1,213
Team bus travel	421	105
Total	**22,694**	**3,756**

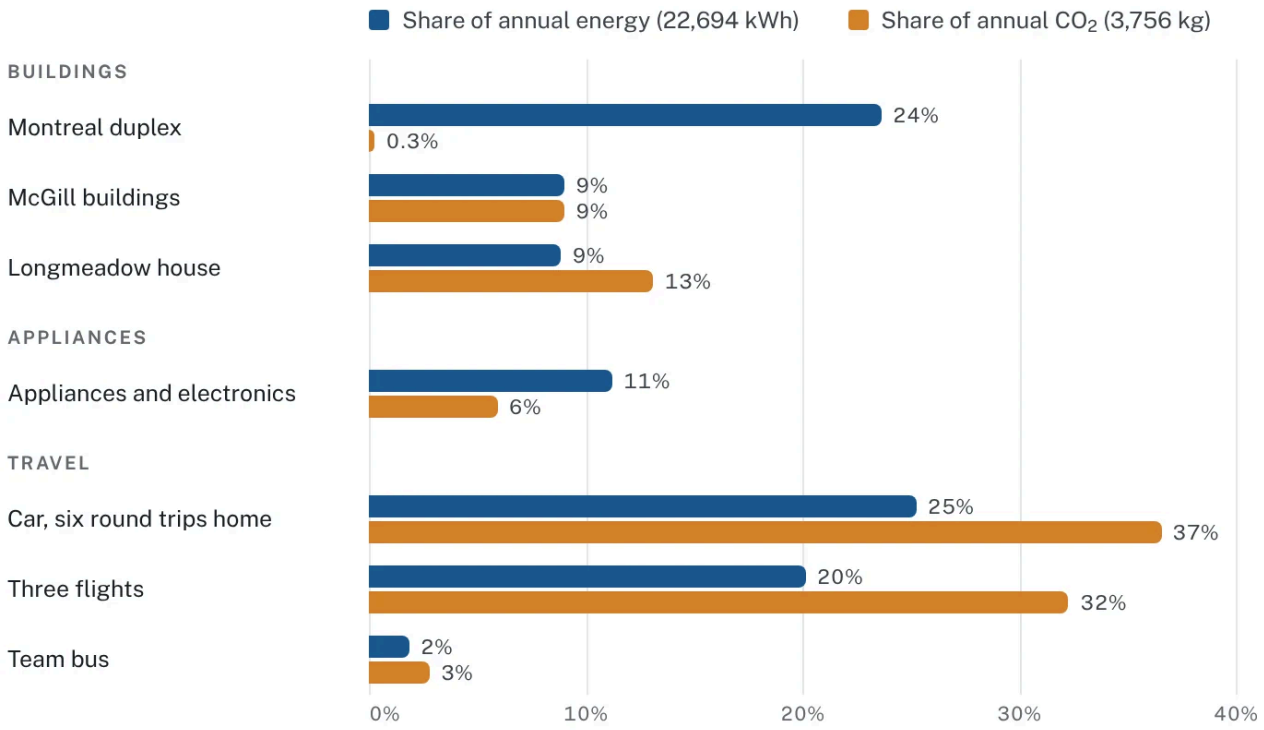


Figure 14.1 Share of annual energy against share of annual CO_2, by source. The Montreal duplex is 24% of the energy and 0.3% of the emissions. The car and the three flights together are 45% of the energy and 69% of the emissions.

Against national figures the total looks small, and the comparison needs care. Canada's total energy supply is 244 kWh per person per day and the audited year runs at about a quarter of it, but national supply includes industry, commerce, freight, energy-sector losses and the energy embodied in goods, none of which appear in a personal audit. The fairer comparison is residential energy, where the Canadian average is 28.6 kWh per person per day and the audited household comes to 27.2. Direct energy alone, at 62.2 kWh a day, is already close to the world's per-capita supply of 58.8.

CARBON FOLLOWS THE GRID, NOT THE KILOWATT-HOUR

The week produced 27.0 kg of CO_2: 16.2 kg from its share of building heating and cooling, mostly McGill's natural gas and the Massachusetts house, 10.8 kg from the Ottawa bus at 4.0 L of diesel per traveller, and 0.09 kg from a week of appliances on Quebec hydroelectricity. Annualized, emissions reach 3.76 t. Long car trips at 1.37 t and three flights at 1.21 t are 69% of the total while being 45% of the energy, and the Montreal duplex, 24% of the energy, is 0.3% of the emissions.

That is well below Quebec's 8.8 t and Canada's 17 t per person, though both of those cover every gas and sector, and below the 14.2 t of fossil CO_2 per person in the U.S. It is already near 80% of the world's 4.6 t, a figure that includes industry and goods where this audit does not. Scaled to a projected 8.3 billion people, 3.76 t each is 31 Gt of CO_2 a year, or 8.52 GtC. At 2.124 GtC per ppm, with land and ocean sinks taking up about half, atmospheric CO_2 would rise by about 2.0 ppm a year, below the 3.73 ppm observed in 2024. The gap comes from Quebec's grid and from what the audit leaves out, food and goods above all.

EMBODIED ENERGY: THE FOOD MATTERS MORE THAN THE DISTANCE

Three foods were traced from production through transport: beef strip loin trucked from Alberta, jasmine rice shipped from Thailand, and potatoes grown in Quebec.

Food and portion	Per portion	Per 1,000 kcal	Fuel per unit of food energy
Beef strip loin, Alberta, 227 g	5.52 kWh	10.7 kWh	9.2
Jasmine rice, Thailand, 100 g dry	0.54 kWh	1.5 kWh	1.3
Potatoes, Quebec, 300 g	0.16 kWh	0.7 kWh	0.6

Per portion the steak takes 34 times the energy of the potatoes, and about 15 times as much per food calorie. Beef needs 9.2 kWh of industrial energy for every kWh of food it provides, while potatoes return more food energy than the fuel spent on them, because sunlight does most of the work. Distance barely moves these numbers. Shipping rice 13,000 km by container adds only 0.20 kWh/kg, and transport is 15% of the beef total against 28% for local potatoes. The calculated beef figure of 88 MJ/kg sits close to the 72 MJ/kg of an independent farm-to-plate estimate, and the same data suggest that switching from beef to chicken would nearly halve life-cycle energy.

COST AND LABOUR

Buying the week's 273 kWh from a single source shows how widely energy is priced. From Hydro-Québec it would cost $25.89, as Énergir natural gas $17.18, as gasoline $56.66, as white rice $326.25, and as AA batteries $48,065. Per kWh that spans more than three orders of magnitude, from 6.3¢ for gas to $176 for batteries, with price rising alongside processing, taxes and portability. Electricity costs more than gas per kWh, but it runs any device and turns into heat with almost no loss at the point of use.

Against human work, a fit labourer sustains about 75 W of mechanical output over an eight-hour shift, 0.6 kWh a day or 3.0 kWh over a five-day week. The audit week equals the output of 91 full-time workers, or 137 at 50 W for an untrained adult. The comparison crosses energy forms, mechanical work against mostly low-temperature heat, and a worker eating 3,000 kcal a day converts only about 17% of it to work.

WHAT THE AUDIT COUNTS, AND WHAT IT LEAVES OUT

This is a direct-energy audit. It leaves out the energy embodied in goods, services and food beyond the three traced, and a summer landscaping job that runs on gas tools. Building heat enters as a modelled annual average rather than a weekly measurement, appliance draws are typical ratings for each model rather than metered values, and several inputs rely on proxies: an Italian rice life-cycle study for Thai jasmine rice, England and Wales data for potatoes, and a name-brand datasheet for store-brand batteries.

The main conclusion does not hinge on those details. Quebec electricity is roughly 130 times cleaner per kWh than New England's, so using less of it in Montreal would save money but barely change emissions. The levers that move the carbon are travel and diet.

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Unlocking Locust Survival: A Chemical Analysis of Locusts

Slow transcriptional control and fast enzymatic control in the same animal. That pairing is what lets a locust change phase in under two hours without changing its DNA.

Koran, F., Leader, C., Stoilova, A., & Weng, A. (2025). Unlocking Locust Survival: A Chemical Analysis of Locusts. *Bioengineering Hyperbook*, Department of Bioengineering, McGill University.

Co-authored chapter published to the departmental Hyperbook.
bioengineering.hyperbook.mcgill.ca/unlocking-locust-survival-a-chemical-analysis-of-locusts

About twenty species in the Acrididae family undergo phase polyphenism. They switch between a solitary phase, where they behave like individual grasshoppers, and a gregarious phase, where they swarm. The trigger is environmental, principally population density and resource availability, and the switch changes colour, behaviour and physiology without altering the genome. The chapter treats that switch as a chemical control system and works through it in three parts: the hormones that set colour, the neurotransmitters and genes that set behaviour, and the pheromones that coordinate the group.

BODY-COLOUR POLYMORPHISM

The life cycle runs from egg pod through five nymphal instars to adult, each instar ended by ecdysis. Juvenile hormone, secreted by the corpora allata into the hemolymph, holds the nymph in its juvenile state by suppressing ecdysone. As its concentration falls, ecdysone gets released and adult features develop. Its structure explains its reach. A terpenoid backbone assembled from three isoprenoid units, a *cis* oxirane ring and a *trans* double bond at C6 let it bind the Methoprene-tolerant protein, which recruits coactivators, moves to the nucleus and acts at gene promoters as inducer or repressor. The ester group before C1 allows hydrolysis, so the signal gets terminated rather than persisting, and that is what makes the timing tunable.

Colour follows from that transcriptional control, but only partly. Solitary locusts are green and earth-brown for background matching; gregarious locusts are black with red, orange and yellow, a warning pattern that reads as unpalatable whether or not the species actually eats toxic flora. Nymphs with elevated juvenile hormone develop green and never show gregarious colouration as adults, which puts juvenile hormone in control of the green pigment alone. Implanting the corpora allata of a black locust into a green one increased black expression, which points at a second, dark-inducing compound from the same gland.

An albino strain was used to isolate it. Treated extracts implanted into albino locusts identified the factor as a heat-stable neuropeptide, which was then matched to [His^7]-corazonin, a hormone previously isolated from a locust with no function assigned to it. Synthetic [His^7]-corazonin reproduced the effect dose-dependently, and the timing of injection determined the outcome. At the start of an instar the locust emerged fully black after ecdysis; mid-instar it developed black patterning on an orange ground; late in the instar it became brown or purple. None of them reproduced a wild gregarious locust exactly, and that is the evidence that both hormones act together.

[His^7]-corazonin works by a different route. Eleven residues long, with histidine at position seven fitting the binding pocket of a G-protein-coupled receptor on epidermal cells, it triggers a conformational change, GDP-for-GTP exchange on the G-protein, and second messengers, cyclic AMP or calcium, that activate the melanin synthesis enzymes directly. Transcriptional control over long periods and enzyme activation over short ones is the contrast the section rests on. It is what allows a rapid phase switch on top of an orderly developmental programme.

BEHAVIOURAL PHASE CHANGE

Two sensory pathways drive the behavioural switch. A thoracic pathway stimulated mechanosensorily by contact between locusts, and a cephalic pathway driven by olfactory and visual cues. Touch, sight and smell are therefore the operative stimuli. The two best-characterized species behave asymmetrically. The migratory locust *Locusta migratoria* gregarizes slowly and solitarizes quickly; the desert locust *Schistocerca gregaria* crowds rapidly and solitarizes slowly.

Dopamine was traced through the catecholamine pathway, the most affected pathway in the fourth stadium of gregarious development. Quantitative PCR on head tissue across five stadia showed the *pale* gene, which encodes the rate-limiting tyrosine hydroxylase, expressed highly in the first stadium and weakening by the last, while *vat1*, which encodes the protein handling synaptic release, rose in the opposite direction. Synthesis early, release late. Both genes were less expressed in solitary locusts. Injection experiments then showed dopamine reducing responsiveness rather than raising it, with 32% of injected gregarious locusts and 20% of injected solitary locusts responding to stimuli against 60% and 38% of saline controls. That is consistent with reduced synaptic efficacy at the thoracic neurons controlling locomotion.

Serotonin was tested in both directions. In the desert locust, receptor antagonists blocked the switch. Gregarization probability fell to 0.27 for the thoracic pathway and 0.07 for the cephalic one. A serotonin precursor produced significant gregarization only when the injected locust was also crowded. So serotonin accelerates the change but does not substitute for the stimulus. In the migratory locust the correlation inverted. Serotonin injection accelerated solitarization, brain serotonin rose during isolation, and 5-HT receptor expression increased during crowding in a way that inhibits the switch. The chapter does not leave that contradiction standing. The two results differ in species, in tissue sampled (thoracic ganglia against brain), in exposure duration (short-term against long-term systemic), and in method (direct injection against gene analysis).

Underneath both neurotransmitters, expression levels of 532 genes differ between phases. Chemosensory protein genes and the *takeout* gene *LmigTO1* separate the phases most sharply, and RNA interference reverses behaviour in each direction: knocking down a chemosensory protein gene moved a gregarious locust toward solitary, and knocking down *takeout* moved a solitary locust toward gregarious. Since neither affects locomotor activity, these genes initiate the transition rather than carry it out.

CHEMICAL COMMUNICATION

Reproduction runs on pheromone chemistry. Females emit sex pheromones, long-chain aldehydes, alcohols and esters built from fatty acids by oxidation, reduction and elongation, from glands near the abdomen. Males detect them on antennal chemoreceptors and respond with wing vibration, antennal movement and copulatory posture, so mate recognition survives the density of a swarm. Cuticular hydrocarbons carry a second layer of information. Produced by fatty acid elongation and modified by methylation or saturation, their profiles shift across the female reproductive cycle and signal receptivity, so males can avoid courting non-receptive females. Females also release courtship-inhibiting pheromones outright, which suppresses male sexual behaviour and keeps control of mating timing on the female side.

Hormones set the schedule for all of it. Juvenile hormone triggers sex pheromone biosynthesis during the fertile phase and raises female receptivity in the same window, and ecdysteroids, which regulate moulting, time reproductive behaviour and the release of both sex and inhibitory pheromones.

At group scale, aggregation pheromones, largely cuticular hydrocarbon derivatives released by both immature and mature locusts, increase locomotion and mutual attraction. Production is density-dependent, so aggregation is self-reinforcing: more locusts means more pheromone means stronger attraction. The same signals hold the group in behavioural synchrony, which is what makes long-distance migration possible, and bring individuals of similar age and reproductive state together, which synchronizes mating. Ovipositing females release a further pheromone that recruits other females to the same egg-laying site, so eggs end up concentrated where hatching succeeds. The froth surrounding the eggs carries water-soluble aggregation factors, which is how phase information passes to the next generation before it hatches. Under crowding or resource scarcity, volatile compounds from injured or dead locusts appear to act as alarm signals and are implicated in cannibalism, on chemical pathways that may overlap with those governing aggregation.

SCOPE

This is a co-authored literature review, not experimental work. Every result cited is another group's, and the chapter's contribution is the synthesis: principally reading juvenile hormone and [His^{7}]-corazonin as one two-timescale control system, and reconciling the opposing serotonin results by species, tissue and method rather than reporting them as a conflict.

Coverage is uneven by necessity, since only a small number of the roughly twenty phase-changing species have been characterized chemically, and most mechanistic work is confined to *L. migratoria* and *S. gregaria*.

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N370S Mutation in Glucocerebrosidase and Type 1 Gaucher Disease

The substitution is not in the active site and it does not collapse the fold. It sits 13 Å away, and that is enough.

Gaucher disease is a lysosomal storage disorder. Glucocerebrosidase normally cleaves glucocerebroside into glucose and ceramide as cells recycle old membranes, and the reticuloendothelial system, spanning bone marrow, spleen and liver, needs a great deal of that activity to clear old blood cells. When the enzyme underperforms, lipid accumulates in macrophage lysosomes and the cells swell into the wrinkled tissue-paper appearance of Gaucher cells. The consequences follow the tissues those cells occupy. Marrow crowding produces anemia and thrombocytopenia, spleen and liver enlarge, and bone pain and fracture come from marrow disruption and inflammatory signalling.

The disease is autosomal recessive and rare, at roughly 1 in 40,000 to 60,000, and carriers of a single variant are typically asymptomatic. Type 1, the non-neuronopathic form, accounts for about 90% of cases, and N370S is the substitution associated with it.

THE WILD TYPE ENZYME

Glucocerebrosidase is a single 497-residue chain in three domains, solved by X-ray crystallography (PDB 1OGS). The catalytic domain is a TIM barrel, eight β-strands in a cylinder wrapped by eight α-helices, which is the standard architecture for enzymes that cut sugars, and it carries the active site. An immunoglobulin-like domain and a β-sheet-rich domain stabilize the enzyme and mediate its interactions. It works at lysosomal pH near 5, is essentially inactive at neutral pH, and depends on saposin C to present the lipid substrate.

WHAT THE MUTANT STRUCTURE SHOWS, AND WHAT IT DOES NOT

The N370S structure (PDB 3KE0) is nearly identical to wild type. The fold is intact, the barrel is intact, and the substitution is not in the active site. That is consistent with the clinical picture, since type 1 patients retain partial enzyme function and a milder course. A structure that looked broken would not explain a disease this variable.

The substitution sits on helix 7, the longest helix in the protein, positioned between the barrel and the immunoglobulin-like domain, and 13 Å from the catalytic centre. Asparagine at that position stabilizes both the local helix geometry and the interface between domains. Serine is smaller, so the hydrogen bonding pattern changes and the helix shifts without unfolding. Hydrogen bonding between the active site region and helix 7 is measurably reduced, and that alone is enough to cost significant activity.

There is a second, compounding effect that is not visible in a solved structure at all. Destabilization early in folding impairs post-ER transport, so less enzyme reaches the lysosome in the first place. The net deficiency is the product of two terms, less enzyme delivered and each molecule less efficient, which is why a substitution this conservative and this far from the active site still produces a storage disease.

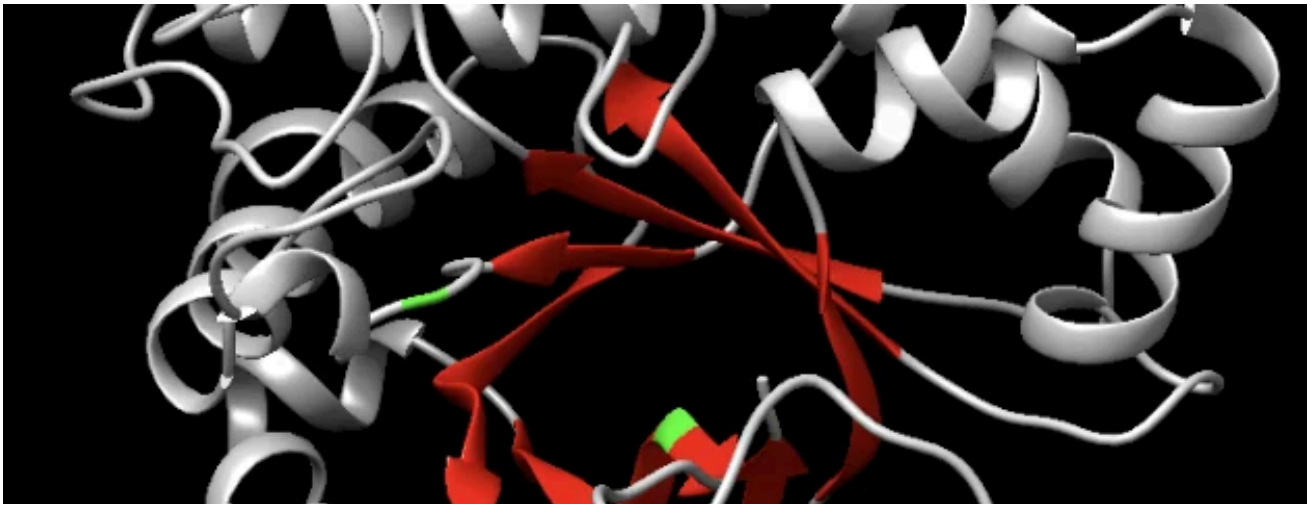


Figure 16.1 Glucocerebrosidase, with the eight-stranded barrel of the catalytic domain in red and the catalytic residues in green. Residue 370 is nowhere near either.

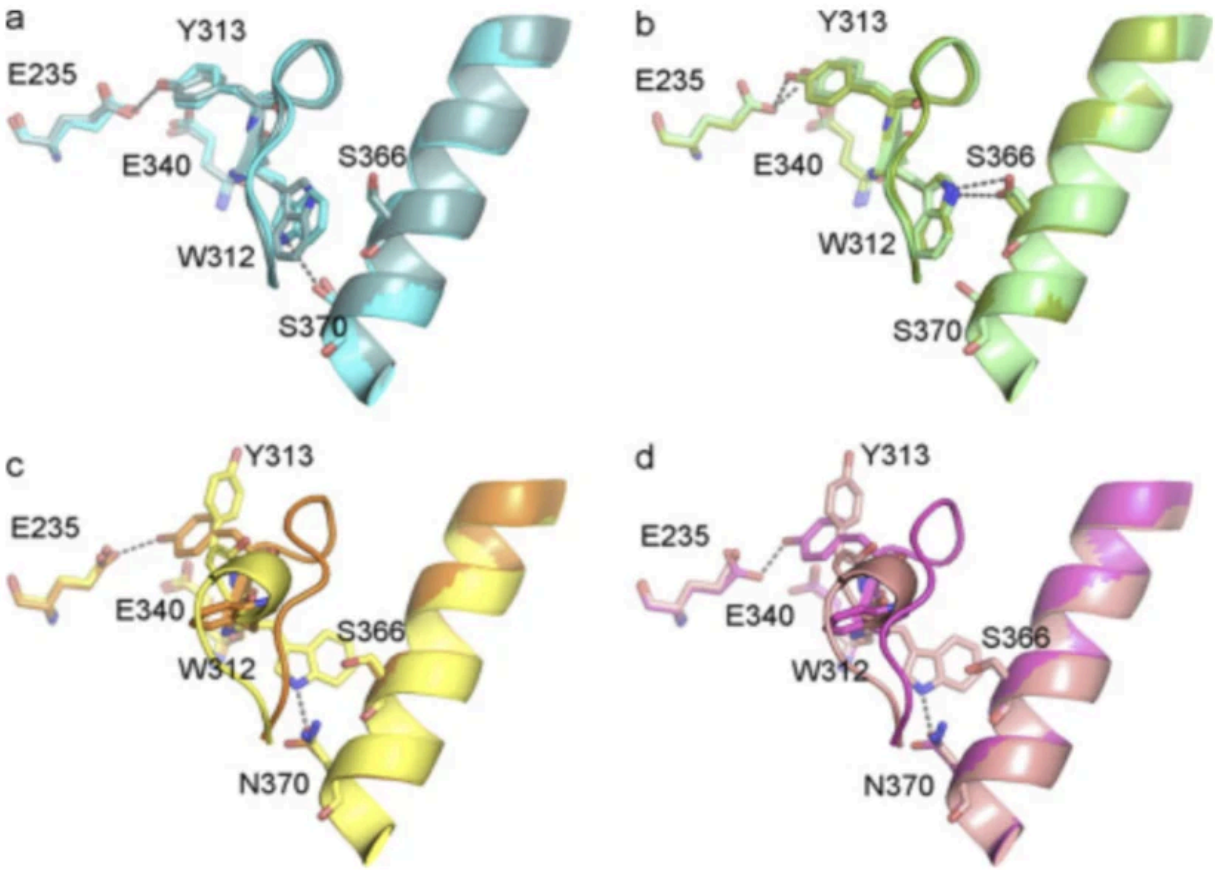


Figure 16.2 Helix 7 in the wild type and N370S structures. Residue 370 sits at the top of the helix, and the hydrogen bonding network around it changes when asparagine is replaced by the smaller serine.

WHY THE TREATMENTS WORK, AND WHAT THAT CONFIRMS

Enzyme replacement therapy infuses functional glucocerebrosidase. Macrophages take it up, stored lipid clears, and blood counts and organ size improve. Substrate reduction therapy attacks the same imbalance from the other side, with oral drugs that lower glucocerebroside production so the remaining enzyme can keep up. Neither is a cure, and lifelong management is the realistic goal, with a near-normal lifespan achievable.

That both approaches work is itself evidence for the mechanism. If the problem were anything other than insufficient lysosomal hydrolase activity, supplying enzyme and reducing substrate would not both resolve the same symptoms.

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Insulin Structural Analysis: 2Zn Pig Insulin at 1.5 Å (PDB 4INS)

Three structural features read off the crystal structure, each traced to the physiological behaviour it produces.

Insulin is small enough to read completely and consequential enough that every structural detail has a clinical readout. This analysis works up the structural hierarchy and then turns around. Three specific features are taken in turn, and each one followed to the behaviour it produces.

THE STRUCTURE, LEVEL BY LEVEL

Level	What is there
Primary	51 amino acids in two chains, A with 21 and B with 30, produced as a single proinsulin precursor and held together after cleavage only by disulfide bonds
Secondary	Two antiparallel α-helices on the A chain at A2–A8 and A12–A18, one α-helix on B at B9–B19, and a β-hairpin from strands at B7–B10 and B20–B23 connecting that helix to the C terminus
Tertiary	A compact globular monomer, locked by three disulfide bridges at A7–B7, A20–B19 and A6–A11, with hydrophobic side chains packed inward and hydrophilic ones outward for solubility in blood
Quaternary	Dimers formed by B chain surfaces meeting in a zipper of hydrogen bonds and hydrophobic contact; three dimers assemble into a hexamer coordinated by zinc, each Zn^{2+} binding HisB10 from three monomers

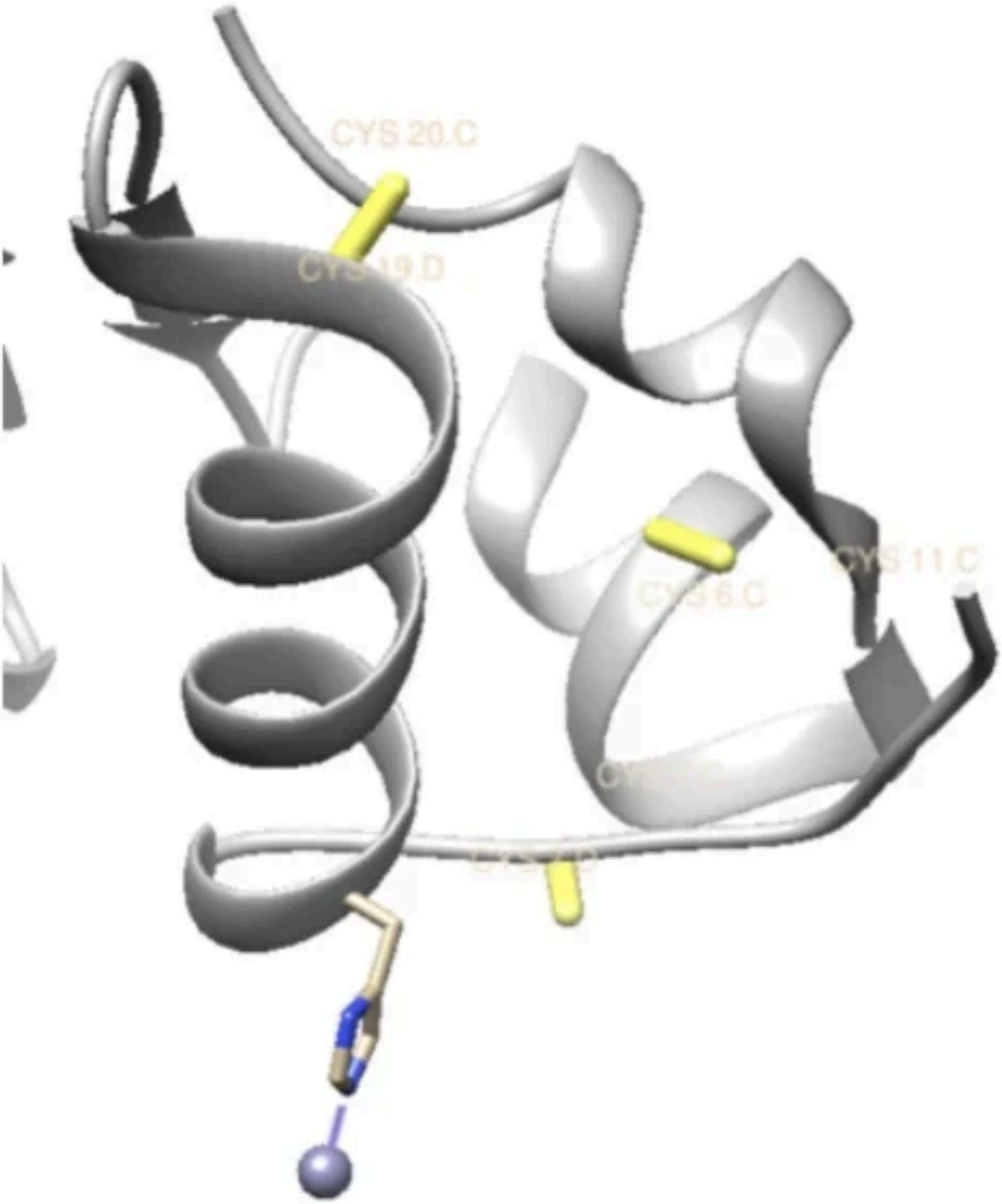


Figure 17.1 The insulin monomer with its three disulfide bridges as yellow sticks. One sits within the A chain; the other two join the A and B chains, which is the only thing holding them together after the proinsulin precursor is cleaved.

THREE FEATURES, AND WHAT EACH ONE BUYS

The disulfide bridges hold the fold together. Two interchain bonds tie A to B and one intrachain bond fixes the A chain's shape. Remove a single one, by mutation or by reduction, and insulin becomes unstable and inactive. The hormone cannot then signal that glucose is high. So the folding requirement is not incidental to function. It is the precondition for it.

The receptor binding surface hides itself until it is needed. Binding uses a pocket of hydrophobic and aromatic residues on B24 to B26, with A3 and A19 contributing, and the critical residue Phe-B24 sits tucked into the hydrophobic core. Its side chain has to swing out before it can contact the receptor. That works as a hinge: insulin stays stable in circulation and exposes its binding face only on arrival, with a secondary site orienting the hormone correctly against the receptor.

The hexamer is a storage format, not a functional one. Only the monomer is active. In the β-cell, zinc assembles insulin into hexamers that pack a high concentration into a small volume without aggregating, stay stable, and dissociate into monomers on demand. The hexamer can also occupy binding sites and slow monomer uptake, which damps rapid swings in blood glucose. Storage, stability and rate control all come out of one assembly step.

FROM RECEPTOR TO PHYSIOLOGY

Insulin binds the α-subunits of its receptor, and the conformational change brings the β-subunits together, which triggers autophosphorylation and then phosphorylation of IRS-1. IRS-1 is the platform the rest of the cascade builds on. Downstream, muscle and fat cells fuse glucose transporters into the membrane and import glucose by facilitated diffusion, the liver converts glucose to glycogen while its glucose-releasing enzymes are switched off, fat breakdown is suppressed, and amino acid uptake and protein synthesis rise. Absent or ineffective insulin, in type 1 deficiency or type 2 resistance, leaves glucose elevated in the blood while the cells that need it are starved.

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Nanopore Signal and Sensing Limits Derived from Geometry and Electrolyte Physics

Three limits on nanopore sequencing, derived rather than measured. Each one trades against another, which is why performance gets quoted as an operating point and not a specification.

A nanopore reads DNA as a current blockade. An electrolyte-filled pore carries an ionic current, a strand passing through displaces ions, and the drop in current is the signal. Everything the technology can and cannot do follows from that geometry. Rather than calibrate the signal empirically, this work derives it, and the derivation puts numbers on where the method runs out.

THE DERIVATION

Open-pore resistance decomposes into two terms: the channel resistance through the pore itself, and the access resistance of the converging ionic paths at each mouth. Both follow from pore geometry and bulk electrolyte conductivity, so the baseline current is predictable from dimensions alone. The occluded case is then expressed as a resistance ratio rather than an absolute value, which makes blockade depth a function of how much of the cross-section the strand occupies. Ion selectivity is handled separately, through Nernst for a single ion and Goldman-Hodgkin-Katz where several ionic species contribute to the potential.

THREE LIMITS, ALL PHYSICAL

The conclusion the derivation supports is that the constraints on nanopore sequencing are physical rather than chemical. No improvement in reagents removes them.

Limit	Magnitude	Origin
Spatial resolution	5 to 6 bases in the sensing window	The constriction is thicker than a single base, so the current reflects a run of them at once
Translocation speed	Microseconds per base unaided, ratcheted to 1 to 100 nucleotides per second	Free electrophoretic transit outruns the amplifier, so an enzyme has to hold the strand back
Noise floor	1/f noise dominant below 1 to 10 kHz	Low-frequency fluctuation sets the smallest blockade distinguishable from baseline

EVERY KNOB TRADES AGAINST ANOTHER

Deriving rather than measuring is what makes the couplings visible. Lowering salt concentration raises blockade contrast, and raises noise with it. Narrowing the pore improves spatial resolution, and shifts the system toward a Debye regime where surface charge governs conduction instead of bulk electrolyte. Speeding the motor enzyme raises throughput, and raises error rate at the same time. No setting improves one term without paying in another. That is why nanopore performance gets stated as an operating point rather than as a specification.

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